

EJERCICIOS RESUELTOS - TP7: POLINOMIO DE TAYLOR Y EXTREMOS

4)

$$a) \quad f(x, y) = f(1, 2) + f'_x(1, 2)(x-1) + f'_y(1, 2)(y-2) + \\ + \frac{1}{2} [f''_{xx}(1, 2)(x-1)^2 + f''_{yy}(1, 2)(y-2)^2 + 2f''_{xy}(1, 2)(x-1)(y-2)] + \\ + \frac{1}{6} [f'''_{xxx}(1, 2)(x-1)^3 + f'''_{yyy}(1, 2)(y-2)^3 + \\ + 3f'''_{xxy}(1, 2)(x-1)^2(y-2) + 3f'''_{xyy}(1, 2)(x-1)(y-2)^2]$$

$$a + b(x-1) + c(y-2) + d(x-1)^2 + e(y-2)^2 + f(x-1)(y-2) + \\ + g(x-1)^3 + h(y-2)^3 =$$

$$a + bx - b + cy - 2c + dx^2 - 2dx + d + ey^2 - 4ey + 4e + \\ + fxy - fy - 2fx + 2f + gx^3 - 3gx^2 + 3gx - g + \\ + hy^3 - 8h - 6hy^2 + 12hy =$$

$$= fxy + (a - b - 2c + d + 4e + 2f - g - 8h) + (b - 2d - 2f + 3g)x + \\ + (d - 3g)x^2 + gx^3 + (c - 4e - f + 12h)y + (e - 6h)y^2 + \\ + hy^3 = x^2 - xy + y^3 - 3 = 0$$

$$\Rightarrow a - b - 2c + d + 4e + 2f - g - 8h = -3 \quad (1)$$

$$b - 2d - 2f + 3g = 0 \quad (2)$$

$$d - 3g = 1 \quad (3)$$

$$g = 0$$

$$f = -1$$

$$c - 4e - f + 12h = 0 \quad (4)$$

$$e - 6h = 0 \quad (5)$$

$$h = 1$$

$$(3) \rightarrow d - 3 \cdot 0 = 1 \Rightarrow d = 1$$

$$(5) \rightarrow e - 6 \cdot 1 = 0 \Rightarrow e = 6$$

$$(4) \rightarrow c - 4 \cdot 6 + 1 + 12 = 0 \Rightarrow c = -13 + 24 \Rightarrow c = 11$$

$$(2) \rightarrow b - 2 + 2 + 0 = 0 \Rightarrow b = 0$$

$$(1) \rightarrow a - 0 - 22 + 1 + 24 - 2 - 0 - 8 = -3 \Rightarrow$$

$$\Rightarrow a = -3 + 8 - 1 \Rightarrow a = 4$$

$$p_1(x,y) = 4 + 11(y-2) + (x-1)^2 + 6(y-2)^2 - (x-1)(y-2) + (y-2)^3$$

b) Si $p(x,y)$ es el polinomio de Taylor de 3° orden, el término a_0 (es decir sin variables x ó y) es

$$f(1,2) =$$

$$\text{Por lo tanto: } f(1,2) = 4$$

Si los polinomios p_0 y p_1 son equivalentes, tenemos que evaluando en $x=1$ y $y=2$ anula todas las potencias de $(x-1)$ y $(y-2)$, quedando el término a_0 . Por ello si hacemos $p(1,2)$ obtenemos el valor dado.

$$p(1,2) = 1^2 - 2 \cdot 1 + 8 - 3 = 1 - 2 + 8 - 3 = 4$$

Sup a) $z - f(x, y) = 0 = F(x, y, z) \quad (\text{see } z_0 = f(x_0, y_0))$

$$\nabla F = (-f'_x, -f'_y, 1) = 0 \nabla F(x_0, y_0, z_0) = \left(-\frac{\partial f}{\partial x}(x_0, y_0), -\frac{\partial f}{\partial y}(x_0, y_0), 1\right)$$

Plane tangent a)

$$\left(-\frac{\partial f}{\partial x}(x_0, y_0), -\frac{\partial f}{\partial y}(x_0, y_0), 1\right) \cdot (x - x_0, y - y_0, z - z_0) = 0 \Rightarrow$$

$$-\frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) - \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) + z - z_0 = 0 \quad (A)$$

Sup b) $z - p(x, y) = 0 = G(x, y, z)$

Sei also $p(x, y) = f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) +$

$$+ \frac{1}{2} \left[\frac{\partial^2 f}{\partial x^2}(x_0, y_0)(x - x_0)^2 + 2 \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0)(x - x_0)(y - y_0) + \right.$$

$$\left. + \frac{\partial^2 f}{\partial y^2}(x_0, y_0)(y - y_0)^2 \right] + \frac{1}{6} [\dots] + \dots + \frac{1}{n!} [\dots]$$

$$p(x, y) = \sum_{i=0}^{\infty} \frac{1}{i!} \left[\frac{\partial}{\partial x}(x - x_0)^i + \frac{\partial}{\partial y}(y - y_0)^i \right]^i (f)$$

$$\nabla G(x, y, z) = (-p'_x, -p'_y, 1) \Rightarrow \nabla G(x_0, y_0, z_0) = (-p'_x(x_0, y_0), -p'_y(x_0, y_0), 1)$$

$$p'_x = \frac{\partial f}{\partial x}(x_0, y_0) + \frac{1}{2} \left[2 \frac{\partial^2 f}{\partial x^2}(x_0, y_0) (x - x_0) + 2 \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) (y - y_0) \right] +$$

$$+ \frac{1}{6} [\dots] + \dots + \frac{1}{n!} [\dots]$$

$$p'_x(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0)$$

Ando lo mismo $p'_y(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0)$

Luego $\rightarrow \nabla G(x_0, y_0, z_0) = \left(-\frac{\partial f}{\partial x}(x_0, y_0), -\frac{\partial f}{\partial y}(x_0, y_0), 1 \right)$

Plano (te b)

$$\left[-\frac{\partial f}{\partial x}(x_0, y_0), -\frac{\partial f}{\partial y}(x_0, y_0), 1 \right] (x - x_0, y - y_0, z - z_0) = 0 \Rightarrow$$

$$\Rightarrow -\frac{\partial f}{\partial x}(x_0, y_0) (x - x_0) - \frac{\partial f}{\partial y}(x_0, y_0) (y - y_0) + z - z_0 = 0 \quad (B)$$

de (A) y (B) $\Rightarrow$ los planos son iguales

$$a) f(2,3) = 2 - 3\sqrt{6-2} = 2 - 3 \cdot 2 = -4$$

$$3) f'_x = 1 + \frac{y}{2\sqrt{6-x}} \Rightarrow f'_x(2,3) = 1 + \frac{3}{4} = \frac{7}{4}$$

$$f'_y = -\sqrt{6-x} \Rightarrow f'_y(2,3) = -2$$

$$f''_{xx} = +\frac{1}{4} y (6-x)^{-3/2} \Rightarrow f''_{xx}(2,3) = \frac{1}{4} \cdot 3 \cdot 4^{-3/2} =$$

$$= \frac{3}{4} \cdot \frac{1}{\sqrt{4^3}} = \frac{3}{32}$$

$$f''_{yy} = 0 \Rightarrow f''_{yy}(2,3) = 0$$

$$f''_{xy} = \frac{1}{2\sqrt{6-x}} \Rightarrow f''_{xy}(2,3) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$p_2(x,y) = -4 + \frac{7}{4}(x-2) - 2(y-3) + \frac{1}{2} \left[\frac{3}{32}(x-2)^2 + \right.$$

$$\left. + 0(y-3)^2 + 2 \cdot \frac{1}{4}(x-2)(y-3) \right]$$

$$p_2(x,y) = -4 + \frac{7}{4}(x-2) - 2(y-3) + \frac{3}{64}(x-2)^2 + \frac{(x-2)(y-3)}{4}$$

$$3) b) f(x, y) = y \ln x + x \ln(y-x) \quad \bar{A} = (1, 2)$$

$$f(1, 2) = 2 \ln 1 + 1 \ln(2-1) = 0$$

$$f'_x = \frac{y}{x} + \ln(y-x) - \frac{x}{y-x} \Rightarrow f'_x(1, 2) = 2 + 0 - \frac{1}{1} = 1$$

$$f'_y = \ln x + \frac{x}{y-x} \Rightarrow f'_y(1, 2) = 0 + \frac{1}{1} = 1$$

$$f''_{xx} = -\frac{y}{x^2} - \frac{1}{y-x} - \left(\frac{y-x+x}{(y-x)^2} \right) \Rightarrow f''_{xx}(1, 2) = -2 - 1 - \left(\frac{2}{1} \right) \\ \Rightarrow f''_{xx}(1, 2) = -5$$

$$f''_{yy} = x \left(-\frac{1}{(y-x)^2} \right) \Rightarrow f''_{yy}(1, 2) = -1$$

$$f''_{xy} = \frac{1}{x} + \left(\frac{y-x+x}{(y-x)^2} \right) \Rightarrow f''_{xy}(1, 2) = 1 + \frac{2}{1} = 3$$

$$P_2(x, y) = 0 + 1(x-1) + 1(y-2) + \frac{1}{2} \left[-5(x-1)^2 + (-1)(y-2)^2 + 2 \cdot 3(x-1)(y-2) \right] \\ = (x-1) + (y-2) + \frac{5}{2}(x-1)^2 - \frac{1}{2}(y-2)^2 + 3(x-1)(y-2)$$

$$c) f(1, 2, 0) = e^0 - (1) \sqrt{1} = 1 - 1 = 0$$

$$3) f'_x = 3e^{x/3} - (3+1) \frac{-1}{2\sqrt{y-x}} \Rightarrow f'_x(1, 2, 0) = -1 \cdot \frac{-1}{2\sqrt{1}} = \frac{1}{2}$$

$$f'_y = -(3+1) \frac{1}{2\sqrt{y-x}} \Rightarrow f'_y(1, 2, 0) = -1 \frac{1}{2\sqrt{1}} = -\frac{1}{2}$$

$$f'_z = x e^{x/3} - \sqrt{y-x} \Rightarrow f'_z(1, 2, 0) = 1 - 1 = 0$$

$$f''_{xx} = 3^2 e^{x/3} + \frac{(3+1)}{2} \left(-\frac{1}{2} \right) (y-x)^{-3/2} (-1) \Rightarrow f''_{xx}(1, 2, 0) = -\frac{1}{2} \left(-\frac{1}{2} \right) 1 = \frac{1}{4}$$

$$f''_{yy} = -\frac{(3+1)}{2} \left(-\frac{1}{2} \right) (y-x)^{-3/2} \Rightarrow f''_{yy}(1, 2, 0) = \left(-\frac{1}{2} \right) \left(-\frac{1}{2} \right) 1 = \frac{1}{4}$$

$$f''_{zz} = x^2 e^{x/3} \Rightarrow f''_{zz}(1, 2, 0) = 1$$

$$f''_{xy} = \frac{3+1}{2} \left(-\frac{1}{2} \right) (y-x)^{-3/2} \Rightarrow f''_{xy}(1, 2, 0) = -\frac{1}{4}$$

$$f''_{xz} = e^{xz} + zxe^{xz} + \frac{1}{2\sqrt{y-x}} \Rightarrow f''_{xz}(1,2,0) = 1 + \frac{1}{2} = \frac{3}{2}$$

$$f''_{yz} = -\frac{1}{2\sqrt{y-x}} \Rightarrow f''_{yz}(1,2,0) = -\frac{1}{2}$$

$$p_2(x,y,z) = 0 + \frac{1}{2}(x-1) - \frac{1}{2}(y-2) + 0 + \frac{1}{2} \left[+\frac{1}{4}(x-1)^2 + \right. \\ \left. +\frac{1}{4}(y-2)^2 + z^2 + 2 \cdot \left(-\frac{1}{4}\right) (x-1)(y-2) + 2 \cdot \frac{3}{2} (x-1)z + \right. \\ \left. + 2 \cdot \left(-\frac{1}{2}\right) (y-2)z \right] =$$

$$= \frac{x}{2} - \frac{1}{2} - \frac{y}{2} + 1 + \frac{1}{8}(x-1)^2 + \frac{1}{8}(y-2)^2 + \frac{3}{2}z - \frac{1}{4}(xy-y- \\ -2x+2) + \frac{3}{2}(xz-z) - \frac{1}{2}(zy-2z)$$

$$3) \downarrow) f(x, y, z) = \cos(x+y) e^{3-x} \quad \bar{A} = (0, 0, 0)$$

$$f(0, 0, 0) = \cos 0 \cdot e^0 = 1$$

$$f'_x = -\sin(x+y) e^{3-x} - e^{3-x} \cos(x+y) \Rightarrow f'_x(0, 0, 0) = -1$$

$$f'_y = -\sin(x+y) e^{3-x} \Rightarrow f'_y(0, 0, 0) = 0$$

$$f'_z = \cos(x+y) \cdot e^{3-x} \Rightarrow f'_z(0, 0, 0) = 1$$

$$f''_{xx} = -e^{3-x} (-\sin(x+y) - \cos(x+y)) + e^{3-x} (-\cos(x+y) + \sin(x+y)) \Rightarrow f''_{xx}(0, 0, 0) = -1(0-1) + 1(-1-0) = 0$$

$$f''_{xy} = -\cos(x+y) e^{3-x} + e^{3-x} \sin(x+y) \Rightarrow f''_{xy}(0, 0, 0) = -1$$

$$f''_{xz} = e^{3-x} (-\sin(x+y) - \cos(x+y)) \Rightarrow f''_{xz}(0, 0, 0) = -1$$

$$f''_{yy} = -\cos(x+y) e^{3-x} \Rightarrow f''_{yy}(0, 0, 0) = -1$$

$$f''_{zz} = e^{3-x} \cos(x+y) \Rightarrow f''_{zz}(0, 0, 0) = 1$$

$$f''_{yz} = -e^{3-x} \sin(x+y) \Rightarrow f''_{yz}(0, 0, 0) = 0$$

$$P_2(x, y, z) = 1 + (-1)x + 0 \cdot y + 1 \cdot z + \frac{1}{2}(-0 \cdot x^2 - 1y^2 + 1z^2 + 2 \cdot (-1)xy + 2 \cdot (-1)xz + 2 \cdot 0 \cdot yz) =$$

$$P_2(x, y, z) = 1 - x + z - \frac{1}{2}y^2 + \frac{1}{2}z^2 - xy - xz$$

4) $0,98^{2,01}$

$f(x, y) = x^y$ $p_0 = (1, 2)$ $dx = -0,02$ $dy = 0,01$

Benardete Taylor hasta 2º orden

$f(1, 2) = 1^2 = 1$

$f'_x = y x^{y-1} \Rightarrow f'_x(1, 2) = 2 \cdot 1^1 = 2$

$f'_y = x^y \ln x \Rightarrow f'_y(1, 2) = 1^2 \ln 1 = 0$

$f''_{xx} = y(y-1) x^{y-2} \Rightarrow f''_{xx}(1, 2) = 2 \cdot 1 \cdot 1^0 = 2$

$f''_{yy} = \ln x \cdot x^y \ln x \Rightarrow f''_{yy}(1, 2) = 0$

$f''_{xy} = x^{y-1} + y \cdot x^{y-1} \ln x \Rightarrow f''_{xy}(1, 2) = 1^2 + 0 = 1$

$f(1 + (-0,02), 2 + 0,01) \approx 1 + 2 \cdot (-0,02) + 0 \cdot 0,01 +$
 $+ \frac{1}{2} [2 \cdot (-0,02)^2 + 0 \cdot (0,01)^2 + 2 \cdot 1 \cdot 0,01 \cdot (-0,02)] =$

$= 1 - 0,04 + 0,0004 - 0,0002 = 0,9602$

$0,98^{2,01} \approx 0,9602$

$$4)b) \sqrt{399} + \sqrt[3]{8,06}$$

$$f(x, y) = \sqrt{x} + \sqrt[3]{y} \quad P = (4, 8) \quad dx = -0,01 \quad dy = 0,06$$

Desarrolla Taylor hasta 2° orden

$$f(4, 8) = \sqrt{4} + \sqrt[3]{8} = 2 + 2 = 4$$

$$f'_x = \frac{1}{2\sqrt{x}} \Rightarrow f'_x(4, 8) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$f'_y = \frac{1}{3} y^{-2/3} \Rightarrow f'_y(4, 8) = \frac{1}{3} \cdot 8^{(-2/3)} = 0,08\bar{3}$$

$$f''_{xx} = \frac{1}{2} \left(-\frac{1}{2}\right) x^{-3/2} \Rightarrow f''_{xx}(4, 8) = -\frac{1}{4} \cdot 4^{-3/2} = -0,03125$$

$$f''_{yy} = \frac{1}{3} \left(-\frac{2}{3}\right) y^{-5/3} \Rightarrow f''_{yy}(4, 8) = -\frac{2}{9} \cdot 8^{-5/3} = 0,0069\bar{4}$$

$$f''_{xy} = 0 \Rightarrow f''_{xy}(4, 8) = 0$$

$$f(4 + (-0,01); 8 + 0,06) \approx 4 + \frac{1}{4}(-0,01) + 0,08\bar{3} \cdot 0,06 + \\ + \frac{1}{2}[-0,03125 \cdot (-0,01)^2 + 0,0069\bar{4} \cdot 0,06^2 + 2 \cdot 0] = 4,00249093$$

$$4)c) \frac{8,97}{302}$$

Desarrolla Taylor hasta 2° orden

$$f(x, y) = \frac{x}{y} \quad P = (9, 3) \quad dx = -0,03 \quad dy = 0,02$$

$$f(9, 3) = \frac{9}{3} = 3$$

$$f'_x = \frac{1}{y} \Rightarrow f'_x(9, 3) = \frac{1}{3}$$

$$f'_y = -\frac{x}{y^2} \Rightarrow f'_y(9, 3) = -\frac{9}{9} = -1$$

$$f''_{xx} = 0 \Rightarrow f''_{xx}(9, 3) = 0$$

$$f''_{yy} = +2 \frac{x}{y^3} \Rightarrow f''_{yy}(9, 3) = \frac{2 \cdot 9}{27} = \frac{2}{3}$$

$$f''_{xy} = -\frac{1}{y^2} \Rightarrow f''_{xy}(9, 3) = -\frac{1}{9}$$

$$f(9 + (-0,03); 3 + 0,02) \approx 3 + \frac{1}{3}(-0,03) - 1 \cdot 0,02 + \\ + \frac{1}{2} \left[0 \cdot (-0,03)^2 + \frac{2}{3} \cdot 0,02^2 + 2 \cdot \left(-\frac{1}{9}\right) (-0,03)(0,02) \right] = \\ = 2,9702$$

05) $f(x, y) = y g(x)$ $P = (x_0, y_0)$ $dx = (x - x_0)$
 desarrollo Taylor hasta 1º orden $dy = (y - y_0)$

$$f(x_0, y_0) = y_0 g(x_0)$$

$$f'_x = y g'(x) \Rightarrow f'_x(x_0, y_0) = y_0 g'(x_0)$$

$$f'_y = g(x) \Rightarrow f'_y(x_0, y_0) = g(x_0)$$

$$\begin{aligned} f(x_0 + (x - x_0), y_0 + (y - y_0)) &= f(x, y) \approx y_0 g(x_0) + \\ &+ y_0 g'(x_0) (x - x_0) + g(x_0) (y - y_0) = \\ &= \cancel{y_0 g(x_0)} + y_0 x g'(x_0) - y_0 x_0 g'(x_0) + g(x_0) y - \cancel{y_0 g(x_0)} = \\ &= g(x_0) y + g'(x_0) y_0 (x - x_0) \quad (A) \end{aligned}$$

Necesito (A) = $g(x_0) (x + y - x_0)$ luego:

$$\cancel{g(x_0) y} + g'(x_0) y_0 (x - x_0) = \cancel{g(x_0) y} + g(x_0) (x - x_0) \Rightarrow$$

$$\Rightarrow g'(x_0) y_0 (x - x_0) = g(x_0) (x - x_0) \quad \text{Si } x \neq x_0$$

En general:

$$g' y_0 = g \Rightarrow \frac{dg}{dx} y_0 = g \Rightarrow \int \frac{dg}{g} y_0 = \int dx$$

$$\Rightarrow \ln k g = x \Rightarrow \ln k g = \frac{x}{y_0} \Rightarrow$$

$$\Rightarrow e^{x/y_0} = k g \Rightarrow g = C e^{x/y_0}$$

$$g(x) = C e^{x/y_0} \quad C \in \mathbb{R}$$

6)a) $f(x, y) = 2 + \sqrt{|xy|}$

$f(0, 0)$ es mínimo absoluto (y relativo)

$$\begin{aligned} \forall (x, y) \in \mathbb{R}^2: \sqrt{|xy|} &\geq 0 \Rightarrow 2 + \sqrt{|xy|} \geq 2 + 0 \Rightarrow \\ &\Rightarrow 2 + \sqrt{|xy|} \geq 2 = f(0, 0) \end{aligned}$$

Luego, por def resulta $f(0, 0)$ mín relat y absoluto

No puede analizarse con la condic. necesaria porque $\nexists f'_x$ y $\nexists f'_y$

$$6)b) f(x,y) = x^3 + xy^2$$

$$f(0,0) = 0$$

$$f'_x = 3x^2 + y^2$$

$$f'_y = 2xy$$

$$f'_x(0,0) = 0$$

$$f'_y(0,0) = 0$$

condic necesaria

$$f''_{xx} = 6x$$

$$f''_{xy} = 2y$$

$$f''_{yy} = 2x$$

$$H(0,0) = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$$

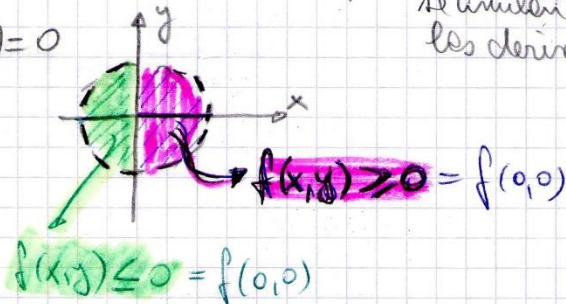
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Comparamos $f(x,y)$ con $f(0,0) = 0$

$$f(x,y) = x(x^2 + y^2)$$

$$\text{si } x > 0 \\ f(x,y) \geq 0$$

$$\text{si } x < 0 \\ f(x,y) \leq 0$$



En cualquier $E(0,0)$ existen puntos (x,y) de modo que:

- $f(x,y) < 0 = f(0,0)$ si $x < 0$

- $f(x,y) > 0 = f(0,0)$ si $x > 0$

por lo tanto $(0,0, f(0,0)) = (0,0,0)$ es pto de silla
dura

b) c)

$$f'_x = -3x^2(x-y^2) + (y-x^3)$$

$$f'_y = (x-y^2) - 2y(y-x^3)$$

$$\left. \begin{array}{l} f'_x(0,0)=0 \\ f'_y(0,0)=0 \end{array} \right\} \Rightarrow (0,0) \text{ es pto crítico de } f$$

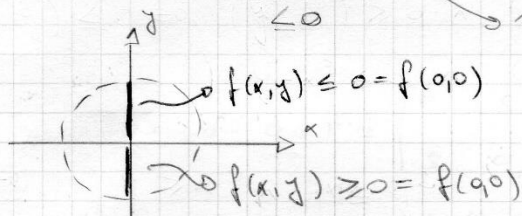
$$f(x,y) = (y-x^3)(x-y^2)$$

$$f(0,0)=0$$

Análisis en $E(0,0)$ -

Nos situamos sobre la recta $x=0$ (eje y) (con $y \neq 0$)

$$f(0,y) = y(-y^2) \begin{cases} \text{si } y > 0 \Rightarrow f(0,y) < 0 \\ \text{si } y < 0 \Rightarrow f(0,y) > 0 \end{cases}$$



En cualquier $E(0,0)$ existen puntos de la forma $(0,y)$ siendo

$$\bullet (0,y) \text{ con } y > 0 \rightarrow f(x,y) < 0 = f(0,0)$$

$$\bullet (0,y) \text{ con } y < 0 \rightarrow f(x,y) > 0 = f(0,0)$$

Luego existen en cualquier $E(0,0)$ puntos para los cuales el valor de la función es mayor que $f(0,0)$ y pts para los cuales es menor que $f(0,0)$ -

Por lo tanto $(0,0, f(0,0)) = (0,0,0)$ es pto de silla dura.

$$7) \quad f(x,y) = (x+y)^4$$

$$\left. \begin{aligned} f'_x &= 4(x+y)^3 = 0 \\ f'_y &= 4(x+y)^3 = 0 \end{aligned} \right\} \Rightarrow (x+y)^3 = 0 \Rightarrow x+y=0 \Rightarrow x=-y$$

$$f''_{xx} = 12(x+y)^2$$

$$f''_{xy} = 12(x+y)^3$$

$$f''_{yy} = 12(x+y)^3$$

$$H(x,y) = \begin{vmatrix} 12(x+y)^2 & 12(x+y)^3 \\ 12(x+y)^3 & 12(x+y)^3 \end{vmatrix} = 0 \quad \forall (x,y) \in \mathbb{R}^2$$

Debo analizar en un entorno del punto

$$\text{Sea } x=-a=0 \quad P=(-a,a) \quad \text{con } a \in \mathbb{R}$$

$$f(-a,a) = 0^4 = 0$$

$$f(x,y) = (x+y)^4 \geq 0 \quad \forall (x,y) \in \mathbb{R}^2 \quad \left. \vphantom{f(x,y)} \right\} \Rightarrow$$

$$\Rightarrow f(x,y) \geq 0 = f(-a,a)$$

Sea $(x,y) \in E(-a,a) : f(-a,a) \leq f(x,y)$ - Por lo tanto, se cumple la def y $f(-a,a)$ es un mínimo local en sentido amplio

08)

$$a) f'_x = 2x + y^2 = 0 \quad (1)$$

$$f'_y = 2y + 2xy = 0 \quad (2)$$

$$(2) \rightarrow 2y(1+x) = 0 \Rightarrow y=0 \vee x=-1$$

$$\bullet y=0 \text{ reemplazando en (1)} \rightarrow 2x=0 \Rightarrow x=0$$

$$P_1 = (0, 0)$$

$$\bullet x=-1 \text{ reemplazando en (1)} \rightarrow -2 + y^2 = 0 \Rightarrow y^2 = 2 \Rightarrow y = \sqrt{2} \vee y = -\sqrt{2}$$

$$P_2 = (-1, \sqrt{2})$$

$$P_3 = (-1, -\sqrt{2})$$

$$f''_{xx} = 2 \quad f''_{yy} = 2 + 2x \quad f''_{xy} = 2y$$

$$H = \begin{vmatrix} 2 & 2y \\ 2y & 2+2x \end{vmatrix}$$

$$P_1 = (0, 0) \quad H(P_1) = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 > 0 \rightarrow \text{extremo local}$$

$$f''_{xx}(0, 0) = 2 > 0 \Rightarrow f(0, 0) \text{ es m\u00ednimo local}$$

$$f(0, 0) = 0$$

$$P_2 = (-1, \sqrt{2}) \quad H(P_2) = \begin{vmatrix} 2 & 2\sqrt{2} \\ 2\sqrt{2} & 2-2 \end{vmatrix} = 0 - 4 \cdot 2 = -8 < 0$$

↓
pto ensillad

$(-1, \sqrt{2}, f(-1, \sqrt{2}))$ es pto ensilladura

$$f(-1, \sqrt{2}) = 1 + 2 + (-1)2 = 1$$

$$P_3 = (-1, -\sqrt{2}) \quad H(P_3) = \begin{vmatrix} 2 & -2\sqrt{2} \\ -2\sqrt{2} & 2-2 \end{vmatrix} = 0 - 4 \cdot 2 = -8 < 0$$

↓
pto ensillad

$(-1, -\sqrt{2}, f(-1, -\sqrt{2}))$ es pto ensilladura

$$f(-1, -\sqrt{2}) = 1 + 2 - 2 = 1$$

$$8)b) \begin{cases} f'_x = 3x^2 - 2x + y = 0 \\ f'_y = x = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ x = 0 \end{cases} \Rightarrow P_1 = (0, 0)$$

$$f''_{xx} = 6x - 2 \quad f''_{yy} = 0 \quad f''_{xy} = 1$$

$$H = \begin{vmatrix} 6x-2 & 1 \\ 1 & 0 \end{vmatrix} \Rightarrow H(P_1) = \begin{vmatrix} -2 & 1 \\ 1 & 0 \end{vmatrix} = -1 < 0 \quad \text{pto ensillado}$$

$(0, 0, f(0, 0)) = (0, 0, 0)$ es pto ensillado

No hay extremos

$$8)c) \text{ Dom } f = \{(x, y) \in \mathbb{R}^2 / y - x > 0\}$$

- Análizo el conj abierto $A = \{(x, y) \in \mathbb{R}^2 / y - x > 0\} \subseteq \text{Dom } f$

$$f'_x = \frac{-1}{2\sqrt{y-x}}$$

$$f'_y = \frac{1}{2\sqrt{y-x}}$$

$$\left. \begin{matrix} f'_x \neq 0 \quad \forall (x, y) \in A \\ f'_y \neq 0 \quad \forall (x, y) \in A \end{matrix} \right\} \Rightarrow \text{no hay extremos sobre } A$$

- Análizo el conj $B = \{(x, y) \in \mathbb{R}^2 / y - x = 0\} \subseteq \text{Dom } f$

$$\left. \begin{matrix} \text{Si } (x_0, y_0) \in B \Rightarrow f(x_0, y_0) = 0 \\ \text{Si } (x_1, y_1) \in A \Rightarrow f(x_1, y_1) > 0 \end{matrix} \right\} \Rightarrow f(x_0, y_0) < f(x_1, y_1)$$

$$\forall (x_0, y_0) \in B \wedge \forall (x_1, y_1) \in A$$

Por lo tanto los pto $(x_0, y_0) \in B$ f alcanza un mínimo absoluto en sentido amplio

$$8)d) \text{ Dom } f = \{(x, y) \in \mathbb{R}^2 / y - x > 0\} \text{ es abierto}$$

$$f'_x = \frac{-1}{y-x} \quad \left. \begin{matrix} f'_x \neq 0 \quad \forall (x, y) \in \text{Dom } f \\ f'_y \neq 0 \quad \forall (x, y) \in \text{Dom } f \end{matrix} \right\}$$

$$f'_y = \frac{1}{y-x} \quad \left. \begin{matrix} f'_y \neq 0 \quad \forall (x, y) \in \text{Dom } f \end{matrix} \right\}$$

No hay extremos en f

$$8) e) 10 - (x-4)^2 - (y-2)^2 \geq 0 \Rightarrow 10 \geq (x-4)^2 + (y-2)^2$$

$$\text{Dom } f = \{(x, y) \in \mathbb{R}^2 / (x-4)^2 + (y-2)^2 \leq 10\}$$

$$\bullet \text{ Analizamos el conj } A = \{(x, y) \in \mathbb{R}^2 / (x-4)^2 + (y-2)^2 < 10\}$$

Siendo A abierto y $A \subseteq \text{Dom } f$

$$f'_x = \frac{-2(x-4)}{2\sqrt{10-(x-4)^2-(y-2)^2}} = 0 \Rightarrow x=4$$

$$f'_y = \frac{-2(y-2)}{2\sqrt{10-(x-4)^2-(y-2)^2}} = 0 \Rightarrow y=2$$

$$P_1 = (4, 2)$$

$$f''_{xx} = \frac{\sqrt{10} - (x-4) \frac{-2(x-4)}{2\sqrt{10-(x-4)^2-(y-2)^2}}}{(\sqrt{10-(x-4)^2-(y-2)^2})^2}$$

$$f''_{yy} = \frac{\sqrt{10} - (y-2) \frac{-2(y-2)}{2\sqrt{10-(x-4)^2-(y-2)^2}}}{(\sqrt{10-(x-4)^2-(y-2)^2})^2}$$

$$f''_{xy} = \frac{0 - \frac{-2(y-2)(x-4)}{2\sqrt{10-(x-4)^2-(y-2)^2}}}{(\sqrt{10-(x-4)^2-(y-2)^2})^2} = -\frac{(y-2)(x-4)}{(10)^{3/2}}$$

$$H(P_1) = \begin{vmatrix} \frac{\sqrt{10}}{(\sqrt{10})^2} & 0 \\ 0 & -\frac{\sqrt{10}}{10} \end{vmatrix} = \frac{(\sqrt{10})^2}{100} > 0 \rightarrow \text{extremo local}$$

$$f''_{xx}(P_1) = \frac{\sqrt{10}}{10} < 0 \rightarrow \text{máximo local}$$

$$f(4, 2) = \sqrt{10} \text{ es máximo local}$$

$$\bullet \text{ Analizamos el conj } B = \{(x, y) \in \mathbb{R}^2 / (x-4)^2 + (y-2)^2 = 10\}$$

Siendo $B \subseteq \text{Dom } f$

$$\left. \begin{array}{l} \text{si } (x_0, y_0) \in B \Rightarrow f(x_0, y_0) = 0 \\ \text{si } (x_1, y_1) \in A \Rightarrow f(x_1, y_1) > 0 \end{array} \right\} \Rightarrow f(x, y_1) > f(x_0, y_0) \\ \forall (x_1, y_1) \in A \wedge \forall (x_0, y_0) \in B$$

Por lo tanto los pts $(x_0, y_0) \in B$, f alcanza un mínimo absoluto

Resumen

$f(4, 2) = \sqrt{10}$ es máximo local y absoluto

$f(x_0, y_0) = 0$ (con $(x_0, y_0) \in B$) son mínimos absolutos en sentido amplio

$$8) f) \begin{cases} f'_x = 2x = 0 \\ f'_y = 4y^3 = 0 \\ f'_z = 3z^2 = 0 \end{cases} \Rightarrow x=y=z=0 \Rightarrow P_1 = (0,0,0)$$

$$f''_{xx} = 2, \quad f''_{yy} = 12y^2, \quad f''_{zz} = 6z$$

$$f''_{xy} = f''_{xz} = f''_{yz} = 0$$

$$H = \begin{vmatrix} 2 & 0 & 0 \\ 0 & 12y^2 & 0 \\ 0 & 0 & 6z \end{vmatrix} \Rightarrow H(P_1) = \begin{vmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} = 0 \rightarrow \text{caso dudoso}$$

Analizamos qué sucede con los valores de f en un entorno de $(0,0,0)$

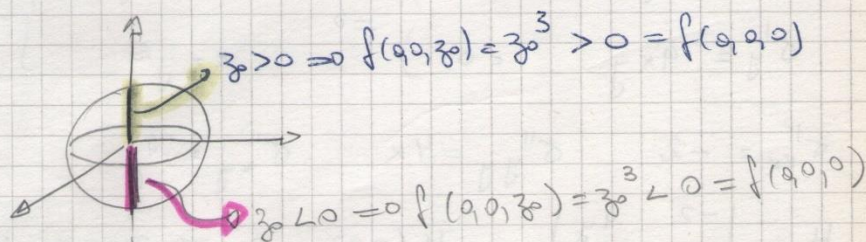
$$f(x,y,z) = x^2 + y^4 + z^3$$

Si evaluo los pts sobre el eje z en $E(\bar{0})$, siendo

$x=y=0$, tenemos que:

$$f(0,0,z_0) = z_0^3 \begin{cases} \rightarrow z_0 > 0 \Rightarrow f(0,0,z_0) > 0 = f(0,0,0) \\ \rightarrow z_0 < 0 \Rightarrow f(0,0,z_0) < 0 = f(0,0,0) \end{cases} \quad (*)$$

Por lo tanto, en $E(0,0,0)$ tenemos pts de la forma $(0,0,z_0)$ donde se verifica $(*)$. Por lo tanto, el pto $(0,0,0, f(0,0,0)) = (0,0,0,0)$ es pto de silla de



9) EXTREMOS EN Puntos INTERIORES

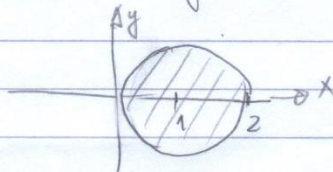
$$f(x,y) = \frac{x}{x^2+y^2+1}$$

$$x^2+y^2-2x \leq 0$$

$$x^2-2x+1+y^2 \leq 1$$

$$(x-1)^2+y^2 \leq 1$$

$$\begin{cases} f'_x = \frac{x^2+y^2+1-2x^2}{(x^2+y^2+1)^2} = \\ = \frac{y^2-x^2+1}{(x^2+y^2+1)^2} = ① \end{cases}$$



$$\begin{cases} f'_y = \frac{-2yx}{(x^2+y^2+1)^2} = ② \end{cases}$$

$$f'_y = 0 \Rightarrow x=0 \vee y=0$$

$$y=0 \text{ en } ① \rightarrow -x^2+1=0 \Rightarrow x^2=1 \begin{matrix} \nearrow x=1 \\ \searrow x=-1 \end{matrix}$$

$$P_1 = (1,0)$$

$$P_2 = (-1,0)$$

no pertenece al recinto

$$f''_{yx} = \frac{-2y(x^2+y^2+1)^2 - 2(x^2+y^2+1)(2x)(-2yx)}{(x^2+y^2+1)^4}$$

$$f''_{yy} = \frac{-2x(x^2+y^2+1)^2 - 2(x^2+y^2+1)(2y)(-2yx)}{(x^2+y^2+1)^4}$$

$$f''_{xx} = \frac{-2x(x^2+y^2+1)^2 - 2(x^2+y^2+1)^2(2x)(y^2-x^2+1)}{(x^2+y^2+1)^4}$$

$$H(1,0) = \begin{vmatrix} \frac{-2 \cdot 4 - 2 \cdot 4 \cdot 2 \cdot 0}{16} & 0 \\ 0 & \frac{-2 \cdot 4}{16} \end{vmatrix} = \begin{vmatrix} -1/2 & 0 \\ 0 & -1/2 \end{vmatrix} = \frac{1}{4} > 0$$

extremo local

NOTA

$$f''_{xx}(1,0) = -1/2 < 0 \rightarrow f(1,0) \text{ max local}$$

9) EN LOS PTS. FRONTERA

HOJA N°

FECHA

$$x^2 + y^2 - 2x = 0$$

$$(x-1)^2 + y^2 = 1$$

$$\begin{cases} x-1 = \cos t \\ y = \sin t \end{cases}$$

Extremos condicionados

$$g(t) = (\cos t + 1, \sin t) \quad 0 \leq t < 2\pi$$

$$f \circ g(t) = f(g(t)) = \frac{\cos t + 1}{(\cos t + 1)^2 + \sin^2 t + 1} =$$

$$= \frac{\cos t + 1}{\cos^2 t + 2\cos t + 1 + \sin^2 t + 1} = \frac{\cos t + 1}{2\cos t + 3}$$

$$h'(t) = \frac{-\sin t (2\cos t + 3) + 2\sin t (\cos t + 1)}{(2\cos t + 3)^2} =$$

$$= \frac{-2\cancel{\sin t} \cos t - 3\sin t + 2\cancel{\sin t} \cos t + 2\sin t}{(2\cos t + 3)^2} =$$

$$= \frac{-\sin t}{(2\cos t + 3)^2} = 0 \Rightarrow t = 0 \vee t = \pi$$

$$t = 0 \rightarrow P_1 = (\cos 0 + 1, \sin 0) = (2, 0)$$

$$t = \pi \rightarrow P_2 = (\cos \pi + 1, \sin \pi) = (0, 0)$$

Clasifico los pts. criticos

$$h''(t) = \frac{-\cos t (2\cos t + 3)^2 - 2(2\cos t + 3)(-2\sin t)(-\sin t)}{(2\cos t + 3)^4}$$

$$h''(0) = \frac{-1(2+3)^2}{(2+3)^4} < 0 \Rightarrow h(0) \text{ es maximo}$$

$$t = 0 \rightarrow P_1 = (2, 0) \quad \left\{ f(2, 0) = \frac{2}{4+1} = \frac{2}{5} \right. \quad \left. \begin{array}{l} \text{m} \acute{o}d \text{ condi} \\ \text{cionado} \end{array} \right.$$

$$h''(\pi) = \frac{-(-1)(-2+3)^2}{(-2+3)^4} > 0 \rightarrow h(\pi) \text{ es m\u00ednimo}$$

$$t = \pi \rightarrow p_2 = (0,0) \quad \boxed{f(0,0) = 0 \text{ es m\u00ednimo condicionado}}$$

Comparo todos los extremos obtenidos

$$f(1,0) = \frac{1}{2} = 0,5 \text{ m\u00e1x local}$$

$$f(2,0) = \frac{2}{5} = 0,4 \text{ es m\u00e1ximo condic}$$

$$f(0,0) = 0 \text{ es m\u00ednimo condic}$$

Luego $f(0,0)$ es m\u00ednimo absoluto

$f(1,0)$ es m\u00e1ximo absoluto

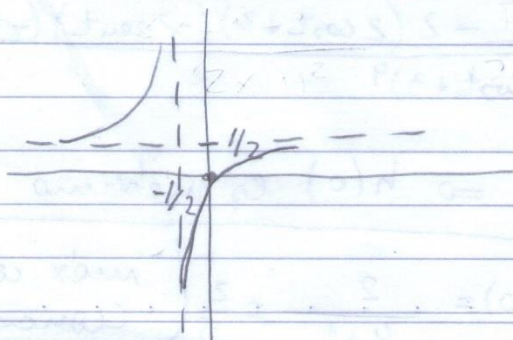
Otra forma

$$x^2 + y^2 = 2x \quad \wedge \quad f(x,y) = \frac{x}{x^2 + y^2 + 1}$$

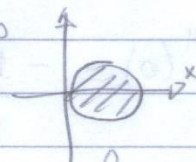
En el conjunto frontera del recinto que se analice:

$$f(x,y) = \frac{x}{x^2 + y^2 + 1}$$

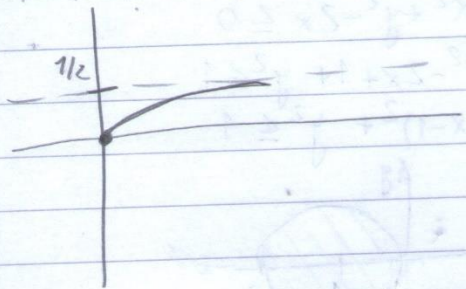
$$g(x) = \frac{x}{2x+1} \text{ homog\u00e9nea}$$



No alcanza extremos, pero en el recinto de analisis



resulta $x \geq 0$ - Luego



$x=0$ es mínimo de $g(x)$
con $x \geq 0$

Lagrange $L(x, y, \lambda) = \frac{x}{x^2 + y^2 + 1} + \lambda(x^2 + y^2 - 2x)$

$$\left\{ \begin{aligned} L'_x &= \frac{y^2 - x^2 + 1}{(x^2 + y^2 + 1)^2} + 2\lambda x = 0 \quad (1) \end{aligned} \right.$$

$$\left\{ \begin{aligned} L'_y &= \frac{-2yx}{(x^2 + y^2 + 1)^2} + 2\lambda y = 0 \quad (2) \end{aligned} \right.$$

$$\left\{ \begin{aligned} L'_\lambda &= x^2 + y^2 - 2x = 0 \quad (3) \end{aligned} \right.$$

SISTEMA COMPLICADO
DE RESOLVER

$$10) \begin{cases} f'_x = 3x^2 = 0 \\ f'_y = 0 \end{cases} \Rightarrow x = 0 \quad P = (0, y) \text{ con } y \in \mathbb{R}$$

Los pts críticos de f son de la forma $P = (0, y)$ con $y \in \mathbb{R}$, es decir, los pts del eje y

Existen infinitos pts (los del eje y) que tienen plano

tge horizontal

$$f''_{xx} = 6x \quad f''_{yy} = 0 \quad f''_{xy} = 0$$

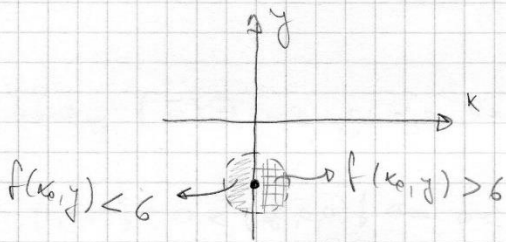
$$H = \begin{vmatrix} 6x & 0 \\ 0 & 0 \end{vmatrix} \Rightarrow H(P) = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = 0 \rightarrow \text{caso dudoso}$$

Análizo los pts en $E(P)$

$$f(x_0, y) = 6 + x_0^3 \begin{cases} \nearrow x_0 > 0 \Rightarrow 6 + x_0^3 > 6 = f(0, y) \\ \searrow x_0 < 0 \Rightarrow 6 + x_0^3 < 6 = f(0, y) \end{cases} \quad (A)$$

Por lo tanto, en el $E(P)$ existen pts para los cuales, f tome valores mayores a $f(0, y) = 6$, y pts para los que tome valores menores a $f(0, y) = 6$ (por A)

Luego el pto $(0, y, 6) = (0, y, f(0, y))$ es un pto de ensilladura



11)

- Analizamos el interior de la placa, es decir que evaluamos los pts críticos en $A = \{(x,y) \in \mathbb{R}^2 / x^2 + y^2 < 4\}$

$$\begin{cases} \sigma'_x = 2y^2 - 2x = 0 \\ \sigma'_y = 4xy = 0 \end{cases} \Rightarrow \begin{cases} x=0 \vee y=0 \\ x=y^2 \end{cases} \Rightarrow \begin{cases} x=0=y=0 \\ x=y^2 \end{cases} \Rightarrow \begin{cases} x=y=0 \\ x=y^2 \end{cases} \Rightarrow p=(0,0)$$

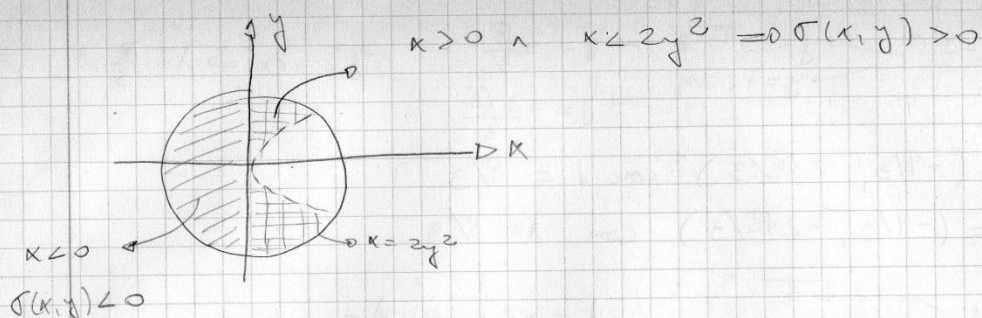
$$\sigma''_{xx} = -2 \quad \sigma''_{yy} = 4x \quad \sigma''_{xy} = 4y$$

$$H = \begin{vmatrix} -2 & 4y \\ 4y & 4x \end{vmatrix} \Rightarrow H(0,0) = 0 \rightarrow \text{caso dudoso}$$

Analizo los valores de σ en un $E(0,0)$

$$\sigma(x,y) = 2xy^2 - x^2 = \underbrace{x}_{<0} \underbrace{(2y^2 - x)}_{>0} \quad \left. \begin{matrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{matrix} \right] <0$$

$$>0 \quad \left. \begin{matrix} 2y^2 - x > 0 \Rightarrow 2y^2 > x \end{matrix} \right] >0$$



En cualquier $E(0)$, existen pts donde:

- i) $x < 0$ ($\because 2y^2 - x > 0$) $\Rightarrow \sigma(x,y) < 0 = \sigma(0,0)$
- ii) $x > 0$ $\wedge$ $x < 2y^2 \Rightarrow \sigma(x,y) = 0$ $\sigma(x,y) > 0 = \sigma(0,0)$

Por i) y ii) podemos afirmar que $(0,0,0)$ es pto de ensi localura

• Analicemos la frontera de la placa, es decir
que eno buemos los pts críticos en $B = \{(x,y) \in \mathbb{R}^2 / x^2 + y^2 = 4\}$

$$L(x,y,\lambda) = 2xy^2 - x^2 + \lambda(x^2 + y^2 - 4)$$

$$L'_x = 2y^2 - 2x + 2\lambda x = 0 \quad (1)$$

$$L'_y = 4xy + 2\lambda y = 0 \quad (2)$$

$$L'_\lambda = x^2 + y^2 - 4 = 0 \quad (3)$$

$$(2) \rightarrow 2y(2x + \lambda) = 0 \Rightarrow y = 0 \vee \lambda = -2x$$

$$y = 0 \mid \text{ en } (3) \rightarrow x^2 = 4 \Rightarrow x = 2 \vee x = -2$$

$$x = 2 \mid \text{ en } (1) \rightarrow 0 - 4 + 4\lambda = 0 \Rightarrow \lambda = 1$$

$$P_1 = (2, 0) \text{ con } \lambda = 1$$

$$x = -2 \mid \text{ en } (1) \rightarrow 0 + 4 - 4\lambda = 0 \Rightarrow \lambda = 1$$

$$P_2 = (-2, 0) \text{ con } \lambda = 1$$

$$\lambda = -2x \mid \text{ en } (1) \rightarrow 2y^2 - 2x + 2(-2x)x = 0 \Rightarrow$$

$$\Rightarrow 2y^2 - 2x - 4x^2 = 0 \Rightarrow y^2 - x - 2x^2 = 0$$

$$\downarrow \text{ en } (3) \mid x = 0 \mid y^2 = 4 - x^2$$

$$\Rightarrow 4 - x^2 - x - 2x^2 = 0 \Rightarrow -3x^2 - x + 4 = 0 \Rightarrow$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 + 48}}{-6}$$

$$\nearrow x = -\frac{4}{3}$$

$$\searrow x = 1$$

Reemp en (3) $\rightarrow |y| = \sqrt{4-x^2}$

$$\bullet x = -\frac{4}{3} \begin{cases} y = \sqrt{4-\frac{16}{9}} \Rightarrow y = \frac{2\sqrt{5}}{3} \\ y = -\sqrt{20/9} \Rightarrow y = -\frac{2\sqrt{5}}{3} \end{cases} \quad \lambda = -2x = 0 \Rightarrow \lambda = \frac{8}{3}$$

$$P_3 = (-4/3, 2\sqrt{5}/3) \text{ con } \lambda = 8/3$$

$$P_4 = (-4/3, -2\sqrt{5}/3) \text{ con } \lambda = 8/3$$

$$\bullet x = 1 \begin{cases} y = \sqrt{3} \\ y = -\sqrt{3} \end{cases} \quad \lambda = -2x = 0 \Rightarrow \lambda = -2$$

$$P_5 = (1, \sqrt{3}) \text{ con } \lambda = -2$$

$$P_6 = (1, -\sqrt{3}) \text{ con } \lambda = -2$$

Clasificamos los pts críticos $P_1, P_2, P_3, P_4, P_5, P_6$

$$i) P_1 = (2, 0) \text{ con } \lambda = 1$$

$$L^* = 2xy^2 - x^2 + x^2 + y^2 - 4 = 2xy^2 + y^2 - 4$$

$$L_x^* = 2y^2 \quad L_y^* = 4xy + 2y \quad g_x' = 2x \quad g_y' = 2y$$

$$L_{xx}^* = 0 \quad L_{yy}^* = 4x + 2 \quad L_{xy}^* = 4y$$

$$H_{\text{del}} L^* = \begin{vmatrix} 0 & 4y & 2x \\ 4y & 4x+2 & 2y \\ 2x & 2y & 0 \end{vmatrix}$$

$$H_{\text{del}}(P_1) = \begin{vmatrix} 0 & 0 & 4 \\ 0 & 10 & 0 \\ 4 & 0 & 0 \end{vmatrix} = 0 + 0 + 4(-40) < 0 \rightarrow \text{mínimo condicionado}$$

$f(2,0) = -4$ es un mínimo condicionado (local)

$$ii) P_2 = (-2, 0) \text{ con } \lambda = 1$$

$$H_{\text{del}} L^* = \begin{vmatrix} 0 & 4y & 2x \\ 4y & 4x+2 & 2y \\ 2x & 2y & 0 \end{vmatrix}$$

$$H_{\text{del}}(P_2) = \begin{vmatrix} 0 & 0 & -4 \\ 0 & -6 & 0 \\ -4 & 0 & 0 \end{vmatrix} = 0 + 0 - 4(-24) > 0 \rightarrow \text{máximo condicionado}$$

$f(-2,0) = -4$ es un máximo condicionado (local)

iii) $P_3 = (-4/3, 2\sqrt{5}/3)$ con $\lambda = 8/3$

$$L^* = 2xy^2 - x^2 + 8/3(x^2 + y^2 - 4)$$

$$L^*_x = 2y^2 - 2x + \frac{16}{3}x \quad L^*_y = 4xy + \frac{16}{3}y \quad g'_x = 2x \quad g'_y = 2y$$

$$L^*_{xx} = -2 + \frac{16}{3} = \frac{10}{3} \quad L^*_{yy} = 4x + \frac{16}{3} \quad L^*_{xy} = 4y$$

$$H_{\text{ord}} L^* = \begin{vmatrix} \frac{10}{3} & 4y & 2x \\ 4y & 4x + \frac{16}{3} & 2y \\ 2x & 2y & 0 \end{vmatrix}$$

$$H_{\text{ord}}(P_3) = \begin{vmatrix} \frac{10}{3} & \frac{8\sqrt{5}}{3} & -\frac{16}{3} \\ -\frac{16}{3} & 0 & \frac{4\sqrt{5}}{3} \\ -\frac{16}{3} & \frac{4\sqrt{5}}{3} & 0 \end{vmatrix} = -\frac{16}{3} \left(\frac{32 \cdot 5}{9} \right) - \frac{4\sqrt{5}}{3} \left(\frac{40\sqrt{5}}{9} - \frac{16^2}{9} \right) =$$

$$= -\frac{2560}{27} + 55,17531885 < 0 \quad \text{mínimo condic}$$

$f(-4/3, 2\sqrt{5}/3) = -\frac{208}{27}$ es mínimo condicionado (local)

iv) $P_4 = (-4/3, -2\sqrt{5}/3)$ con $\lambda = 8/3$

$$H_{\text{ord}}(P_4) = \begin{vmatrix} \frac{10}{3} & -\frac{8\sqrt{5}}{3} & -\frac{8}{3} \\ -\frac{8\sqrt{5}}{3} & -\frac{16}{3} + \frac{16}{3} & -\frac{4\sqrt{5}}{3} \\ -\frac{8}{3} & -\frac{8\sqrt{5}}{3} & 0 \end{vmatrix} = \frac{10}{3} \left(-\frac{32 \cdot 3}{9} \right) + \frac{8\sqrt{5}}{3} \left(\frac{32\sqrt{5}}{9} \right) - \frac{8}{3} \frac{64 \cdot 3}{9} =$$

$$< 0 \rightarrow \text{mínimo condic}$$

$f(-4/3, -2\sqrt{5}/3) = -\frac{8}{3} \cdot \frac{4 \cdot 5}{9} - \frac{16}{9} = -\frac{208}{27}$ es mínimo condic

v) $P_5 = (-1, \sqrt{3})$ con $\lambda = -2$

$$H_{\text{ord}} L^* = \begin{vmatrix} -6 & 4y & 2x \\ 4y & 4x+4 & 2y \\ 2x & 2y & 0 \end{vmatrix}$$

$$L^*_{xx} = -2 - 4 = -6$$

$$L^*_{yy} = 4x - 4$$

$$L^*_{xy} = 4y$$

$$H_{\text{ord}}(P_5) = \begin{vmatrix} -6 & 4\sqrt{3} & 2 \\ 4\sqrt{3} & 0 & 2\sqrt{3} \\ 2 & 2\sqrt{3} & 0 \end{vmatrix} = -6(-4 \cdot 3) - 4\sqrt{3}(-4\sqrt{3}) + 2 \cdot$$

$$\cdot 8 \cdot 3 =$$

$$= 72 + 48 + 48 > 0 \rightarrow \text{máximo condic}$$

$f(-1, \sqrt{3}) = 2 \cdot 3 - 1 = 5$ es máx condic (local)

$$n). p_6 = (1, -\sqrt{3}) \quad \text{con } 1 = -2$$

$$Hess(p_6) = \begin{vmatrix} -6 & -4\sqrt{3} & 2 \\ -4\sqrt{3} & 0 & -2\sqrt{3} \\ 2 & -2\sqrt{3} & 0 \end{vmatrix} = -6(-4 \cdot 3) + 4\sqrt{3}(4\sqrt{3}) + 2 \cdot 8 \cdot 3 =$$

$$= 72 + 48 + 48 > 0 \rightarrow \text{máx condic}$$

$$f(1, -\sqrt{3}) = 2 \cdot 3 - 1 = 5 \text{ es máx condic (local)}$$

Resumen

$$f(2, 0) = -4$$

mínimo condic.

$$f(-2, 0) = -4$$

máximo condic

$$f(-4/3, 2\sqrt{3}/3) = -\frac{208}{27}$$

mínimo condic

$$f(-4/3, -2\sqrt{3}/3) = -\frac{208}{27}$$

mínimo condic

mínimos absolutos

$$f(1, \sqrt{3}) = 5$$

máximo condic

$$f(1, -\sqrt{3}) = 5$$

máximo condic

máximos absolutos

Vol Taylor

$$P(x,y) = 7x + y + xy - y^2 - 4x^2$$

$$12) f'_x(1,1) = P'_x(1,1) \Rightarrow f'_x(1,1) = (7 + y - 8x) \Big|_{(1,1)} = 0$$

$$f'_y(1,1) = P'_y(1,1) \Rightarrow f'_y(1,1) = (1 + x - 2y) \Big|_{(1,1)} = 0$$

Luego $A = (1,1)$ es pto crítico

$$f''_{xx}(1,1) = P''_{xx}(1,1) \Rightarrow f''_{xx}(1,1) = -8$$

$$f''_{yy}(1,1) = P''_{yy}(1,1) \Rightarrow f''_{yy}(1,1) = -2$$

$$f''_{xy}(1,1) = P''_{xy}(1,1) \Rightarrow f''_{xy}(1,1) = 1$$

$$H(A) = \begin{vmatrix} -8 & 1 \\ 1 & -2 \end{vmatrix} = 16 - 1 = 15 > 0 \rightarrow \text{extremo local}$$

$$f''_{xx}(A) = -8 < 0 \rightarrow \text{máx local}$$

$$f(1,1) = P(1,1) = 7 + 1 + 1 - 1 - 4 = 4 \quad \text{es máx local}$$

14)

$$z'_x = f'_x = -\frac{F'_x}{F'_z} \quad z'_y = f'_y = -\frac{F'_y}{F'_z}$$

Se (x_0, y_0) é pts. estacionário de $f = 0$ $\begin{cases} f'_x(x_0, y_0) = 0 \\ f'_y(x_0, y_0) = 0 \end{cases}$

$$z''_{xx} = f''_{xx} = \frac{\partial}{\partial x} \left(-\frac{F'_x}{F'_z} \right) = -\frac{(F''_{xx} + F'_{xz} z'_x) F'_z - (F'_{zx} + F''_{zz} z'_x) F'_x}{(F'_z)^2}$$

$$F'_x \begin{matrix} \nearrow x \\ \rightarrow y \\ \searrow z \end{matrix}$$

analogamente F'_y e F'_z

$$z''_{xx}(x_0, y_0) = f''_{xx}(x_0, y_0) = \frac{-[F''_{xx}(\bar{A}) + F'_{xz}(\bar{A}) z'_x(x_0, y_0)] F'_z(\bar{A}) - [F'_{zx}(\bar{A}) + F''_{zz}(\bar{A}) z'_x(x_0, y_0)] F'_x(\bar{A})}{[F'_z(\bar{A})]^2}$$

$$= \frac{-[F''_{xx}(\bar{A}) + F'_{xz}(\bar{A}) z'_x(x_0, y_0)] F'_z(\bar{A}) - [F'_{zx}(\bar{A}) + F''_{zz}(\bar{A}) z'_x(x_0, y_0)] F'_x(\bar{A})}{[F'_z(\bar{A})]^2}$$

$$= \frac{-F''_{xx}(\bar{A}) F'_z(\bar{A})}{[F'_z(\bar{A})]^2} - \frac{F'_{xz}(\bar{A}) F'_x(\bar{A})}{[F'_z(\bar{A})]^2}$$

→ como $F'_x(\bar{A}) = 0$
 por ser
 $f'_x = -\frac{F'_x}{F'_z} = 0$

$$f''_{xx}(x_0, y_0) = -\frac{F''_{xx}(\bar{A})}{F'_z(\bar{A})}$$

$$f''_{xy} = z''_{xy} = \frac{\partial}{\partial y} (z'_x) = \frac{\partial}{\partial y} \left(-\frac{F'_x}{F'_z} \right) = -\frac{[(F''_{xy} + F'_{xz} z'_y) F'_z - (F'_{zy} + F''_{zz} z'_y) F'_x]}{(F'_z)^2}$$

$$f''_{xy}(x_0, y_0) = \frac{-[F'_{xy}(\bar{A}) + F'_{xz}(\bar{A}) z'_y(x_0, y_0)] F'_z(\bar{A}) - [F'_{zy}(\bar{A}) + F'_{zz}(\bar{A}) z'_y(x_0, y_0)] F'_x(\bar{A})}{[F'_z(\bar{A})]^2}$$

$$= -\frac{F'_{xy}(\bar{A}) F'_z(\bar{A})}{[F'_z(\bar{A})]^2} + \frac{F'_{zy}(\bar{A}) F'_x(\bar{A})}{[F'_z(\bar{A})]^2}$$

→ como $F'_x(\bar{A}) = 0$
 por ser
 $f'_x = -\frac{F'_x}{F'_z} = 0$

$$f''_{xy}(x_0, y_0) = -\frac{F'_{xy}(\bar{A})}{F'_z(\bar{A})}$$

$$P''_{yy} = \frac{\partial}{\partial y} \left(-\frac{F'_y}{F'_z} \right) = - \left[\frac{(F''_{yy} + F''_{yz} \frac{\partial}{\partial y}) F'_z - (F''_{zy} + F''_{zz} \frac{\partial}{\partial y}) F'_y}{(F'_z)^2} \right]$$

$$P''_{yy}(x_0, y_0) = - \frac{(F''_{yy}(\bar{A}) + F''_{yz}(\bar{A}) \frac{\partial}{\partial y}(x_0, y_0)) F'_z(\bar{A}) - (F''_{zy}(\bar{A}) + F''_{zz}(\bar{A}) \frac{\partial}{\partial y}(x_0, y_0)) F'_y(\bar{A})}{[F'_z(\bar{A})]^2}$$

$$= - \frac{F''_{yy}(\bar{A}) F'_z(\bar{A})}{[F'_z(\bar{A})]^2} + \frac{F''_{yz}(\bar{A}) F'_y(\bar{A})}{[F'_z(\bar{A})]^2} \quad \begin{array}{l} \text{wegen } F'_y(\bar{A}) = 0 \\ \text{oder} \\ \frac{\partial}{\partial y} = -\frac{F'_z}{F'_y} = 0 \end{array}$$

$$\boxed{P''_{yy}(x_0, y_0) = - \frac{F''_{yy}(\bar{A})}{F'_z(\bar{A})}}$$

$$15) F(x, y, z) = x^2 + 4y^2 + z^2 - 36 = 0 \text{ p.o. } x=0 \wedge y=0 \Rightarrow z=6 \vee z=-6$$

i) F continua por ser polinómica

$$ii) F(0, 0, 6) = F(0, 0, -6) = 0$$

$$iii) \begin{cases} F'_x = 2x \\ F'_y = 8y \\ F'_z = 2z \end{cases} \left\{ \begin{array}{l} \text{continuas por ser polinómicas} \\ \text{y existen en } E(0, 0, 6) \text{ y } E(0, 0, -6) \end{array} \right.$$

$$iv_A) F'_z(0, 0, 6) = 12 \neq 0$$

Pues queda definida $\varphi_1(x, y)$ en $E(0, 0, 6)$

$$iv_B) F'_z(0, 0, -6) = -12 \neq 0$$

Pues queda definida $\varphi_2(x, y)$ en $E(0, 0, -6)$

A) $\rightarrow \varphi_1$ analizo la existencia de extremos en (0,0)

$$\left. \begin{aligned} \varphi'_1 x &= -\frac{F'_x}{F'_z} = -\frac{2x}{2z} = -\frac{x}{z} = 0 \\ \varphi'_1 y &= -\frac{F'_y}{F'_z} = -\frac{8y}{2z} = -\frac{4y}{z} = 0 \end{aligned} \right\} \Rightarrow x=y=0$$

$P=(0,0)$ es pto crítico de φ_1

$$\varphi''_{xx} = \frac{\partial}{\partial x} \left(-\frac{x}{z} \right) = -\left[\frac{z - z'_x x}{z^2} \right] \Rightarrow \varphi''_{xx}(0,0) = -\frac{6}{36} = -\frac{1}{6}$$

$$\varphi''_{yy} = \frac{\partial}{\partial y} \left(-\frac{4y}{z} \right) = -\left[\frac{4z - z'_y \cdot 4y}{z^2} \right] \Rightarrow \varphi''_{yy}(0,0) = -\frac{\frac{24}{3}}{36} = -\frac{2}{3}$$

$$\varphi''_{xy} = \frac{\partial}{\partial x} \left(-\frac{4y}{z} \right) = -\left[\frac{-z'_x \cdot 4y}{z^2} \right] \Rightarrow \varphi''_{xy}(0,0) = 0$$

$$H(P) = \begin{vmatrix} -1/6 & 0 \\ 0 & -2/3 \end{vmatrix} = \frac{2}{18} > 0 \rightarrow \text{extremo local}$$

$$\varphi''_{xx}(0,0) = -\frac{1}{6} < 0$$

$\varphi_1(0,0)=6$ es máximo local

L_{máximo}
local

B) $\rightarrow \varphi_2$ | analizo la existencia de extremos en $(0,0)$

$$\left. \begin{aligned} \varphi'_{2x} &= -\frac{F'_x}{F'_z} = -\frac{2x}{2z} = -\frac{x}{z} = 0 \\ \varphi'_{2y} &= -\frac{4y}{z} = 0 \end{aligned} \right\} \Rightarrow x=y=0$$

$P = (0,0)$ es pts críticos de φ_2

$$\varphi''_{xx} = -\frac{z - 3x \cdot x}{z^2} \Rightarrow \varphi''_{xx}(0,0) = -\frac{-6}{36} = \frac{1}{6}$$

$$\varphi''_{yy} = -\frac{4z - 3y \cdot 4y}{z^2} \Rightarrow \varphi''_{yy}(0,0) = -\frac{-24}{36} = \frac{2}{3}$$

$$\varphi''_{xy} = -\frac{3x \cdot 4y}{z^2} \Rightarrow \varphi''_{xy}(0,0) = 0$$

$$H^*(P) = \begin{vmatrix} 1/6 & 0 \\ 0 & 2/3 \end{vmatrix} = \frac{1}{6} \cdot \frac{2}{3} > 0 \rightarrow \text{extremo local}$$

$$\varphi''_{xx}(0,0) = \frac{1}{6} > 0 \rightarrow \text{mínimo local}$$

$$\varphi_2(0,0) = -6 \quad \text{es mínimo local}$$

Hallamos las funciones analíticamente

$$x^2 + 4y^2 + z^2 = 36 \Rightarrow z^2 = 36 - x^2 - 4y^2 \Rightarrow$$

$$\Rightarrow |z| = \sqrt{36 - x^2 - 4y^2} \quad \begin{cases} z = \sqrt{36 - x^2 - 4y^2} \\ z = -\sqrt{36 - x^2 - 4y^2} \end{cases}$$

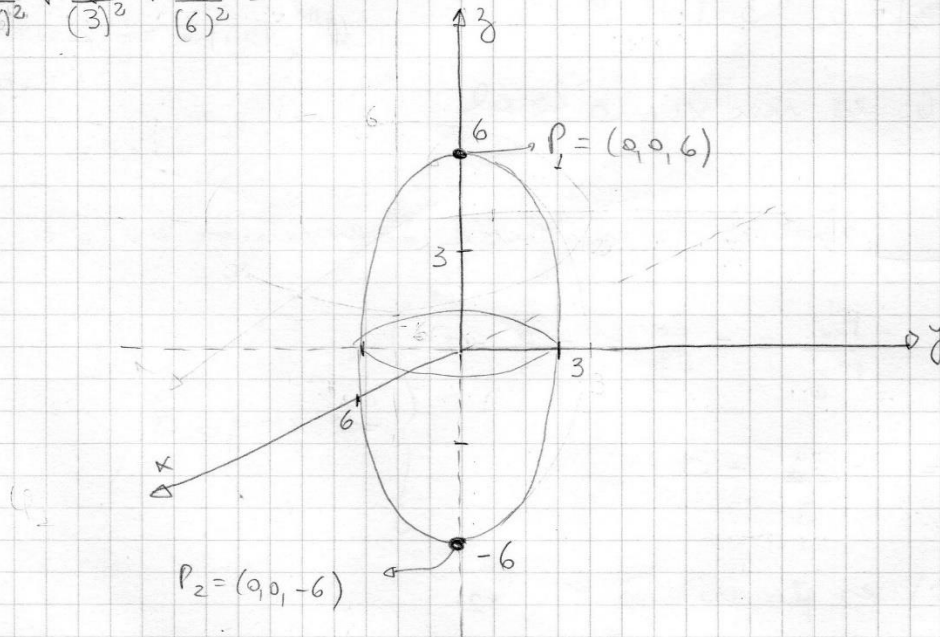
$$\varphi_1(x,y) = \sqrt{36 - x^2 - 4y^2} \rightarrow (0,0,6) \in \text{graf } \varphi_1$$

$$\varphi_2(x,y) = -\sqrt{36 - x^2 - 4y^2} \rightarrow (0,0,-6) \in \text{graf } \varphi_2$$

● Interpretación geométrica

$x^2 + 4y^2 + z^2 = 36$ es un elipsoide

$$\frac{x^2}{(6)^2} + \frac{y^2}{(3)^2} + \frac{z^2}{(6)^2} = 1$$



graf φ_1 es la gráfica de la parte superior del elipsoide

graf φ_2 es la gráfica de la parte inferior del elipsoide

$P_1 = (0, 0, 6) \therefore 6 = \varphi_1(0, 0)$ es máximo de φ_1

$P_2 = (0, 0, -6) \therefore -6 = \varphi_2(0, 0)$ es mínimo de φ_2

$$16) \quad x^2 + 4y^2 + 9z^2 = 1 \quad \bar{A} \in S$$

a) Sea $F(x, y, z) = x^2 + 4y^2 + 9z^2 - 1 = 0$

- F continua en $E(\bar{A})$ por polinómica

- $F(\bar{A}) = 0$ (porque $\bar{A} \in S$)

- $F'_x = 2x$

$F'_y = 8y$

$F'_z = 18z$

} continua en $E(\bar{A})$ por polinómica

- $F'_z = 18z \neq 0$ si $z \neq 0$

Analizo los pts $\bar{A} = (x_0, y_0, z_0)$ con $z_0 \neq 0$

En un entorno de dichos pts, queda definida una $z = f(x, y)$. Para dicho f , analizaré el signo de $H(\bar{A})$

$$z'_x = - \frac{F'_x}{F'_z} = - \frac{2x}{18z} = - \frac{1}{9} \frac{x}{z} = 0 \Rightarrow x = 0$$

$$z'_y = - \frac{F'_y}{F'_z} = - \frac{8y}{18z} = - \frac{4}{9} \frac{y}{z} = 0 \Rightarrow y = 0$$

$$z''_{xx} = - \frac{1}{9} \left(z - \frac{z'_x x}{z^2} \right) = - \frac{1}{9} \left(\frac{z + \frac{1}{9} \frac{x^2}{z}}{z^2} \right) = P = (0, 0)$$

$$= - \frac{1}{9} \left(\frac{9z^2 + x^2/z}{9z^2} \right) = \frac{-9z^2 - x^2}{81z^3}$$

$$z''_{yy} = - \frac{4}{9} \left(z - \frac{z'_y y}{z^2} \right) = - \frac{4}{9} \left(\frac{z + \frac{4}{9} \frac{y^2}{z}}{z^2} \right) = - \frac{4}{9} \cdot \left(\frac{9z^2 + 4y^2}{9z^3} \right)$$

$$= \frac{-36z^2 - 16y^2}{81z^3}$$

$$z''_{xy} = \frac{\partial}{\partial y} \left(- \frac{1}{9} \frac{x}{z} \right) = - \frac{1}{9} x \left(- \frac{1}{z^2} \cdot z'_y \right) = \frac{x}{9z^2} \cdot \left(- \frac{4}{9} \right) \frac{y}{z} =$$

$$= - \frac{4xy}{81z^3}$$

$$H = \begin{vmatrix} \frac{-9z^2 - x^2}{81z^3} & -\frac{4xy}{81z^3} \\ -\frac{4xy}{81z^3} & \frac{-36z^2 - 16y^2}{81z^3} \end{vmatrix} =$$

$$= \frac{-9z^2 - x^2}{81z^3} \left(\frac{-36z^2 - 16y^2}{81z^3} \right) - \left(\frac{4xy}{81z^3} \right)^2 = \frac{9 \cdot 36 z^4 + 9 \cdot 16 z^2 y^2 +}{81^2 z^6}$$

$$+ 36 x^2 z^2 + 16 x^2 y^2 - 16 x^2 y^2 = \frac{9 \cdot 36 z^4 + 9 \cdot 16 z^2 y^2 + 36 x^2 z^2}{81^2 z^6} > 0$$

$$= \frac{9 \cdot 36 z^4 + 9 \cdot 16 z^2 y^2 + 36 x^2 z^2}{81^2 z^6} > 0 \text{ porque es cociente de n}^\circ \text{ positivos (el nume}$$

rador es suma de potencias pares)

A su vez, el numerador nunca se anula porque siendo $z \neq 0$, aunque sean $x=y=0$, termino

$$\text{quedando } \frac{9 \cdot 36 z^4}{81^2 z^6} > 0$$

Luego $z(0,0) = \phi(0,0)$ es un extremo relativo

$$\text{Calificamos el extremo: } z''_{xx}(0,0) = \frac{-9z^2 - 0}{81z^3} =$$

$$= \frac{-9z^2}{81z^3} = -\frac{1}{9z}$$

$$\text{Si } x=0 \text{ y } y=0 \Rightarrow 0^2 + 4 \cdot 0^2 + 9z^2 = 1 \Rightarrow z^2 = \frac{1}{9} \Rightarrow$$

$$\Rightarrow |z| = \frac{1}{3} \Rightarrow z = \frac{1}{3} \vee z = -\frac{1}{3}$$

$$\text{Si } z = \frac{1}{3} \Rightarrow z''_{xx}(\infty) = -\frac{1}{27} < 0 \Rightarrow \phi(\infty) \text{ es m}^\circ \text{ relat}$$

$$\text{Si } z = -\frac{1}{3} \Rightarrow z''_{xx}(\infty) = \frac{1}{27} > 0 \Rightarrow \phi(\infty) \text{ es m}^\circ \text{ relat}$$

$$\phi \text{ no es } \text{ } \begin{cases} \phi_1 = \sqrt{\frac{1-x^2-4y^2}{9}} \\ \phi_2 = -\sqrt{\frac{1-x^2-4y^2}{9}} \end{cases}$$

$$f(x, y) = 4x^3 + 4y^3 + \frac{12}{x} + \frac{12}{y}$$

$$17) a) \begin{cases} z'_x = 12x^2 - \frac{12}{x^2} = 0 & (1) \\ z'_y = 12y^2 - \frac{12}{y^2} = 0 & (2) \end{cases}$$

$$(1) \Rightarrow \frac{12x^4 - 12}{x^2} = 0 \Rightarrow x^4 - 1 = 0 \Rightarrow x^4 = 1 \Rightarrow$$

$$\Rightarrow x = 1 \vee x = -1$$

$$(2) \Rightarrow \frac{12y^4 - 12}{y^2} = 0 \Rightarrow y^4 - 1 = 0 \Rightarrow y = 1 \vee y = -1$$

$$P_1 = (1, 1)$$

$$P_2 = (1, -1)$$

$$P_3 = (-1, 1)$$

$$P_4 = (-1, -1)$$

$$z''_{xx} = 24x - 12(-2)x^{-3} = 24x + \frac{24}{x^3}$$

$$z''_{yy} = 24y + \frac{24}{y^3}$$

$$z''_{xy} = 0$$

$$H = \begin{vmatrix} 24x + \frac{24}{x^3} & 0 \\ 0 & 24y + \frac{24}{y^3} \end{vmatrix}$$

$$H(P_1) = \begin{vmatrix} 48 & 0 \\ 0 & 48 \end{vmatrix} = 48^2 > 0 \rightarrow \text{extremo local}$$

$$f''_{xx}(P_1) = 48 > 0 \rightarrow \text{mínimo local}$$

$$f(1,1) = 8 + 24 = 32 \text{ es m\u00ednimo local}$$

$$H(P_2) = \begin{vmatrix} 48 & 0 \\ 0 & -48 \end{vmatrix} = -48^2 < 0 \rightarrow \text{pto ensilladura}$$

$$(1, -1, f(1, -1)) = (1, -1, 0) \text{ es pto ensilladura}$$

$$H(P_3) = \begin{vmatrix} -48 & 0 \\ 0 & 48 \end{vmatrix} = -48^2 < 0 \rightarrow \text{pto ensilladura}$$

$$(-1, 1, f(-1, 1)) = (-1, 1, 0) \text{ es pto de ensilladura}$$

$$H(P_4) = \begin{vmatrix} -48 & 0 \\ 0 & -48 \end{vmatrix} = 48^2 > 0 \rightarrow \text{extremo local}$$

$$f''_{xx}(P_4) = -48 < 0 \rightarrow \text{m\u00e1ximo local}$$

$$f(-1, -1) = -8 - 24 = -32 \text{ es m\u00e1ximo local}$$

$$7b) f(x,y) = x - y^2 - x^3 + 2xy$$

$$\begin{cases} f'_x = 1 - 3x^2 + 2y = 0 & (1) \\ f'_y = -2y + 2x = 0 & (2) \end{cases}$$

$$(2) \rightarrow 2x = 2y \Rightarrow x = y \quad \text{Reins en (1)} : 1 - 3x^2 + 2x = 0 \Rightarrow \\ \Rightarrow 3x^2 - 2x - 1 = 0$$

$$x_{1,2} = \frac{2 \pm \sqrt{4 + 12}}{6} = \frac{2 \pm 4}{6} \begin{matrix} \nearrow 1 \\ \searrow -\frac{1}{3} \end{matrix}$$

$$x = 1 \Rightarrow y = 1 \quad P_1 = (1, 1)$$

$$x = -\frac{1}{3} \Rightarrow y = -\frac{1}{3} \quad P_2 = (-\frac{1}{3}, -\frac{1}{3})$$

$$f''_{xx} = -6x \quad f''_{yy} = -2 \quad f''_{xy} = 2$$

$$H(1,1) = \begin{vmatrix} -6 & 2 \\ 2 & -2 \end{vmatrix} = 12 - 4 > 0 \quad \text{extremo local}$$

$$f''_{xx}(1,1) = -6 < 0 \rightarrow \text{máx local}$$

$$\boxed{f(1,1) = 1 \text{ es máx local}}$$

$$H(-\frac{1}{3}, -\frac{1}{3}) = \begin{vmatrix} 2 & 2 \\ 2 & -2 \end{vmatrix} = -4 - 4 = -8 < 0 \rightarrow \text{pto silla}$$

$$\boxed{(-\frac{1}{3}, -\frac{1}{3}, f(-\frac{1}{3}, -\frac{1}{3})) \text{ es pto silla}}$$

$$17) c) z'_x = 4(2-4x+3y)^3 (-4) = 0 \quad ①$$

$$z'_y = 4(2-4x+3y)^3 \cdot 3 = 0 \quad ②$$

$$\begin{cases} ① \Rightarrow 2-4x+3y = 0 \\ ② \Rightarrow 2-4x+3y = 0 \end{cases} \Rightarrow y = \frac{-2+4x}{3}$$

$$P = \left(x_0, \frac{4x_0-2}{3}\right) \text{ pts críticos}$$

$$z''_{xx} = -16 \cdot 3(2-4x+3y)^2 (-4)$$

$$z''_{yy} = 12 \cdot 3(2-4x+3y)^2 \cdot 3$$

$$z''_{xy} = -16 \cdot 3(2-4x+3y)^2 \cdot 3$$

$$z''_{xx}(P) = z''_{yy}(P) = z''_{xy}(P) = 0 \quad \therefore \nexists H(P)$$

Analizo los valores de f en $E(P)$

$$f(P) = 0$$

$$(2-4x+3y)^4 \geq 0 \quad \forall (x,y) \in \mathbb{R}^2$$

Por lo tanto $f(P) = 0$ es un mínimo local en sentido amplio

$$17) d) F(x, y, z) = xy - z \cos(yz) + 1 = 0$$

$$z'_x = -\frac{F'_x}{F'_z} = -\frac{y}{-\cos(yz) + zy \sin(yz)} = 0 \quad (1)$$

$$z'_y = -\frac{F'_y}{F'_z} = -\frac{x + z^2 \sin(yz)}{-\cos(yz) + zy \sin(yz)} = 0 \quad (2)$$

$$(1) \Rightarrow y = 0$$

$$(2) \Rightarrow x + z^2 \sin 0 = 0 \Rightarrow x = 0$$

$$P = (0, 0)$$

Aplicando ejercicio 09):

$$z''_{xx}(0,0) = -\frac{F''_{xx}(0,0,z_0)}{F'_z(0,0,z_0)} = 0$$

$$\begin{aligned} z''_{yy}(0,0) &= -\frac{F''_{yy}(0,0,z_0)}{F'_z(0,0,z_0)} = -\frac{z^3 \cos(yz) \big|_{(0,0,z_0)}}{-\cos(yz) + zy \sin(yz) \big|_{(0,0,z_0)}} \\ &= -\frac{z_0^3 \cancel{\cos z_0}}{-\cancel{\cos z_0}} = -z_0^3 \end{aligned}$$

$$\text{Cálculo aux} \rightarrow F(0,0,z_0) = 0 - z_0 \cos z_0 + 1 = 0 \Rightarrow$$

$$\Rightarrow z_0 \cos z_0 = 1 \Rightarrow z_0 = ???$$

$$z''_{xy}(0,0) = -\frac{F''_{xy}(0,0,z_0)}{F'_z(0,0,z_0)} = -\frac{1}{-\cos z_0} = \frac{1}{\cos z_0}$$

$$H(0,0) = \begin{vmatrix} 0 & \frac{1}{\cos z_0} \\ \frac{1}{\cos z_0} & -z_0^3 \end{vmatrix} = -\frac{1}{\cos^2 z_0} < 0 \quad \text{pto silla}$$

$(0,0, f(0,0)) = (0,0,z_0)$ es pto silla local

$$17) e) \begin{cases} x = u \\ y = u - r \\ z = r^2 \end{cases} \Rightarrow f(x, y, z) = u(u - r) + u(r^2) = u^2 - ur + ur^2$$

$$g(u, r) = u^2 - ur + ur^2$$

$$g'_u = 2u - r + r^2 = 0 \quad (1)$$

$$g'_r = -u + 2ur = 0 \quad (2)$$

$$(2) \Rightarrow u(-1 + 2r) = 0 \Rightarrow u = 0 \vee r = \frac{1}{2}$$

$$\bullet u = 0 \text{ en } (1) \Rightarrow r^2 - r = 0 \Rightarrow r(r - 1) = 0 \Rightarrow r = 0 \vee r = 1$$

$$p_1 = (0, 0)$$

$$p_2 = (0, 1)$$

$$\bullet r = \frac{1}{2} \text{ en } (1) \Rightarrow 2u - \frac{1}{2} + \frac{1}{4} = 0 \Rightarrow u = \left(\frac{1}{2} - \frac{1}{4}\right) \frac{1}{2} \Rightarrow$$

$$\Rightarrow u = \frac{1}{8}$$

$$p_3 = \left(\frac{1}{8}, \frac{1}{2}\right)$$

$$g''_{uu} = 2$$

$$g''_{rr} = 2u$$

$$g''_{ur} = -1 + 2r$$

$$H = \begin{vmatrix} 2 & -1+2r \\ -1+2r & 2u \end{vmatrix}$$

$$\bullet H(P_1) = \begin{vmatrix} 2 & -1 \\ -1 & 0 \end{vmatrix} = -1 < 0 \rightarrow \text{pto silla de punto}$$

$$\left. \begin{matrix} u=0 \\ r=0 \end{matrix} \right\} \Rightarrow (x, y, z) = (0, 0, 0)$$

$$(0, 0, 0, f(0, 0, 0)) = (0, 0, 0, 0) \text{ es pto silla de punto}$$

$$\bullet H(P_2) = \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -1 < 0 \rightarrow \text{pto silla de punto}$$

$$\left. \begin{matrix} u=0 \\ r=1 \end{matrix} \right\} \Rightarrow (x, y, z) = (0, -1, 1)$$

$$(0, -1, 1, f(0, -1, 1)) = (0, -1, 1, 0) \text{ es pto silla de punto}$$

$$\bullet H(P_3) = \begin{vmatrix} 2 & -1+2 \cdot \frac{1}{2} \\ -1+2 \cdot \frac{1}{2} & 2 \cdot \frac{1}{8} \end{vmatrix} = 1/2 > 0 \rightarrow \text{extremo local}$$

$$g''_{uu}(P_3) = 2 > 0 \rightarrow \text{mínimo local}$$

$$\left. \begin{matrix} u = \frac{1}{8} \\ r = \frac{1}{2} \end{matrix} \right\} \Rightarrow (x, y, z) = \left(\frac{1}{8}, \frac{1}{8} - \frac{1}{2}, \frac{1}{4} \right) = \left(\frac{1}{8}, -\frac{3}{8}, \frac{1}{4} \right)$$

$$f\left(\frac{1}{8}, -\frac{3}{8}, \frac{1}{4}\right) = \frac{1}{8} \left(-\frac{3}{8}\right) + \frac{1}{8} \cdot \frac{1}{4} = -\frac{1}{64} \text{ es mínimo local}$$

17)

$$f) \begin{cases} x = 2t \\ y = 4t \\ z = t \end{cases} \Rightarrow f(x, y, z) = t^2(2t+9) - 6 \cdot 4t$$

$$g(t) = 2t^3 + 9t^2 - 24t$$

$$g'(t) = 6t^2 + 18t - 24 = 0 \Rightarrow t^2 + 3t - 4 = 0 \Rightarrow t = \frac{-3 \pm \sqrt{9+16}}{2}$$

$$t_1 = -4, t_2 = 1$$

$$g''(t) = 12t + 18$$

$$g''(-4) = -30 < 0 \rightarrow \text{máximo local}$$

$$\bullet t = -4 \Rightarrow (x, y, z) = (-8, -16, -4)$$

$$f(-8, -16, -4) = 16(-8+9) - 6 \cdot (-16) = 112 \text{ es máximo local}$$

$$\bullet g''(-1) = -12 + 18 > 0 \rightarrow \text{mínimo local}$$

$$\bullet t = 1 \Rightarrow (x, y, z) = (2, 4, 1)$$

$$f(2, 4, 1) = 1(2+9) - 6 \cdot 4 = -13 \text{ es mínimo local}$$

$$18) d = \sqrt{(x-1)^2 + (y-7)^2}$$

Maximizar o minimizar
la raíz es equivalente a
maximizar o minimizar su
argumento

$$L(x, y, \lambda) = (x-1)^2 + (y-7)^2 + \lambda(4y^2 - 18x + 9x^2 - 16y - 11)$$

$$L'_x = 2(x-1) - 18\lambda + 18\lambda x = 0 \rightarrow \textcircled{1}$$

$$L'_y = 2(y-7) + 8\lambda y - 16\lambda = 0 \rightarrow \textcircled{2}$$

$$L'_\lambda = 4y^2 - 18x + 9x^2 - 16y - 11 = 0 \textcircled{3}$$

$$\textcircled{1} \rightarrow x-1-9\lambda+9\lambda x=0 \Rightarrow x(1+9\lambda)-(1+9\lambda)=0 \Rightarrow$$

$$\Rightarrow (x-1)(1+9\lambda)=0 \Rightarrow x=1 \vee \lambda=-\frac{1}{9}$$

$$x=1 \text{ en } \textcircled{3} \rightarrow 4y^2 - 18 + 9 - 16y - 11 = 0 \Rightarrow$$

$$\Rightarrow 4y^2 - 16y - 20 = 0 \Rightarrow y^2 - 4y - 5 = 0 \Rightarrow$$

$$\Rightarrow y = \frac{4 \pm \sqrt{16 + 20}}{2} \Rightarrow y = \frac{4 \pm 6}{2} \begin{cases} y=5 \\ y=-1 \end{cases}$$

$$x=1 \wedge y=5 \text{ en } \textcircled{2} \rightarrow 2(5-7) + 8\lambda \cdot 5 - 16\lambda = 0 \Rightarrow$$

$$\Rightarrow -4 + 40\lambda - 16\lambda = 0 \Rightarrow \lambda = \frac{4}{24}$$

24

$$\bullet P_1 = (1, 5) \text{ con } \lambda = \frac{4}{24}$$

24

$$x=1 \wedge y=-1 \text{ en } \textcircled{2} \rightarrow 2(-1-7) - 8\lambda - 16\lambda = 0 \Rightarrow$$

$$\Rightarrow -16 - 24\lambda = 0 \Rightarrow \lambda = -\frac{16}{24} \Rightarrow \lambda = -\frac{2}{3}$$

$$\bullet P_2 = (1, -1) \text{ con } \lambda = -\frac{2}{3}$$

$$\lambda = -1/9 \text{ en (2)} \rightarrow 2(y-7) + 8\left(-\frac{1}{9}\right)y - 16\left(-\frac{1}{9}\right) = 0 \Rightarrow$$

$$\Rightarrow 2y - 14 - \frac{8}{9}y + \frac{16}{9} = 0 \Rightarrow \frac{10}{9}y = \frac{110}{9} \Rightarrow$$

$$\Rightarrow y = 11$$

$$\text{Reemp en (3)} \rightarrow 4 \cdot 11^2 - 18x + 9x^2 - 16 \cdot 11 - 11 = 0 \Rightarrow$$

$$\Rightarrow 9x^2 - 18x + 297 = 0$$

$$x^2 - 2x + 33 = 0 \quad x_{1,2} = \frac{2 \pm \sqrt{4 - 4 \cdot 33}}{2} \notin \mathbb{R}!!$$

clasificamos los pts críticos

$$-P_1 = (1, 5) \text{ con } \lambda = 4/24$$

$$L^* = 2(x-1)^2 + (y-5)^2 + \frac{4}{24} (4y^2 - 18x + 9x^2 - 16y - 11)$$

$$L^*_{xx} = 2 + 18 \cdot \frac{4}{24} = 5$$

$$L^*_{yy} = 2 + 8 \cdot \frac{4}{24} = 10/3$$

$$L^*_{xy} = 0$$

$$H_{\text{al}} = \begin{vmatrix} 5 & 0 & -18+18x \\ 0 & 10/3 & 8y-16 \\ -18+18x & 8y-16 & 0 \end{vmatrix}$$

$$H_{\text{al}}(P_1) = \begin{vmatrix} 5 & 0 & 0 \\ 0 & 10/3 & 24 \\ 0 & 24 & 0 \end{vmatrix} = 5 \begin{vmatrix} 0 & 24^2 \\ 0 & 0 \end{vmatrix} < 0$$

mínimo condic

$$\text{dist}(1,5) = \sqrt{0^2 + (-2)^2} = 2 \text{ es mínima}$$

El punto más cercano es $(1,5)$ y su distancia al pto. dado es 2

$$- p_2 = (1, -1) \text{ con } \lambda = -\frac{2}{3}$$

$$L^* = (x-1)^2 + (y-2)^2 - \frac{2}{3} (4y^2 - 18x + 9x^2 - 16y - 11)$$

$$L^*_{xx} = 2 + 18 \cdot \left(-\frac{2}{3}\right) = -10$$

$$L^*_{yy} = 2 + 8 \left(-\frac{2}{3}\right) = -\frac{10}{3}$$

$$L^*_{xy} = 0$$

$$H_{\text{al}} = \begin{vmatrix} -10 & 0 & -18+18x \\ 0 & -\frac{10}{3} & 8y-16 \\ -18+18x & 8y-16 & 0 \end{vmatrix}$$

$$H_{\text{al}}(p_2) = \begin{vmatrix} -10 & 0 & 0 \\ 0 & -\frac{10}{3} & -24 \\ 0 & -24 & 0 \end{vmatrix} = -10(-24^2) > 0$$

máximo.
condic

$$\text{dist}(1,-1) = \sqrt{0^2 + (-8)^2} = 8 \text{ es máxima}$$

El punto más lejano es $(1,-1)$ y su distancia al pto. dado es 8

$$19) Vol = \pi r^2 h$$

$$Area = 2\pi r \cdot h + 2\pi r^2$$

$$a) L = \pi r^2 h + \lambda (2\pi r h + 2\pi r^2 - A)$$

$$L'_r = 2\pi r h + 2\pi \lambda h + 4\pi \lambda r = 0 \quad (1)$$

$$L'_h = \pi r^2 + 2\pi \lambda r = 0 \quad (2)$$

$$L'_\lambda = 2\pi r h + 2\pi r^2 - A = 0 \quad (3)$$

$$De (2) \rightarrow \pi r (r + 2\lambda) = 0 \Rightarrow r = 0 \vee r = -2\lambda$$

$$i) r = 0 \rightarrow \text{reemp en (3)} : -A = 0 \text{ absurdo}$$

$$ii) r = -2\lambda \text{ reemp en (1)} \rightarrow \cancel{2\pi} r h + \cancel{2\pi} \lambda (-\frac{r}{2}) h + \cancel{4\pi} (-\frac{r}{2}) r = 0$$

$$\Rightarrow r h - \frac{r h}{2} - r^2 = 0 \Rightarrow \frac{r h}{2} - r^2 = 0 \Rightarrow$$

$$\Rightarrow r \left(\frac{h}{2} - r \right) = 0 \Rightarrow r = 0 \text{ (absurdo)} \vee r = \frac{h}{2}$$

$$r = -2\lambda \wedge r = \frac{h}{2} \text{ reemp en (3)} \rightarrow \cancel{2\pi} \frac{h}{2} h + \cancel{2\pi} \frac{h^2}{4} - A = 0$$

$$\Rightarrow \pi h^2 + \pi \frac{h^2}{2} - A = 0 \Rightarrow \frac{3\pi h^2}{2} = A \Rightarrow$$

$$\Rightarrow h^2 = \frac{2A}{3\pi} \Rightarrow h = \sqrt{\frac{2A}{3\pi}} \vee h = -\sqrt{\frac{2A}{3\pi}} \text{ (absurdo)}$$

porque h es
altura del
cilindro

$$P = \left(\frac{1}{2} \sqrt{\frac{2A}{3\pi}}, \sqrt{\frac{2A}{3\pi}} \right) \text{ con } \lambda = +\frac{1}{4} \sqrt{\frac{2A}{3\pi}}$$

Clasifico el pto critico

$$L''_{rr} = 2\pi h + 4\pi \left(-\frac{1}{4} \sqrt{\frac{2A}{3\pi}} \right) \quad L''_{hh} = 0$$

$$L^*_{rh} = 2\pi r + 2\pi \left(-\frac{1}{4} \sqrt{\frac{2A}{3\pi}} \right)$$

$$L^*_{rh}(p) = 2\pi \sqrt{\frac{2A}{3\pi}} + \pi \sqrt{\frac{2A}{3\pi}} = \pi \sqrt{\frac{2A}{3\pi}}$$

$$L^*_{rh}(p) = \cancel{2\pi} \frac{1}{\cancel{2}} \sqrt{\frac{2A}{3\pi}} - \frac{\pi}{2} \sqrt{\frac{2A}{3\pi}} = \frac{\pi}{2} \sqrt{\frac{2A}{3\pi}}$$

$$g(r, h) = 2\pi r h + 2\pi r^2 - A = 0$$

$$g'_r = 2\pi h + 4\pi r \Rightarrow g'_r(p) = 2\pi \sqrt{\frac{2A}{3\pi}} + \cancel{4\pi} \frac{1}{\cancel{2}} \sqrt{\frac{2A}{3\pi}} = 4\pi \sqrt{\frac{2A}{3\pi}}$$

$$g'_h = 2\pi r \Rightarrow g'_h(p) = \cancel{2\pi} \frac{1}{\cancel{2}} \sqrt{\frac{2A}{3\pi}} = \pi \sqrt{\frac{2A}{3\pi}}$$

$$Hess(p) = \begin{vmatrix} \pi \sqrt{\frac{2A}{3\pi}} & \frac{\pi}{2} \sqrt{\frac{2A}{3\pi}} & 4\pi \sqrt{\frac{2A}{3\pi}} \\ \frac{\pi}{2} \sqrt{\frac{2A}{3\pi}} & 0 & \pi \sqrt{\frac{2A}{3\pi}} \\ 4\pi \sqrt{\frac{2A}{3\pi}} & \pi \sqrt{\frac{2A}{3\pi}} & 0 \end{vmatrix} =$$

$$= \pi \sqrt{\frac{2A}{3\pi}} \left(0 - \pi^2 \frac{2A}{3\pi} \right) - \frac{\pi}{2} \sqrt{\frac{2A}{3\pi}} \left(-4\pi^2 \frac{2A}{3\pi} \right) + 4\pi \sqrt{\frac{2A}{3\pi}} \cdot$$

$$\cdot \pi^2 \frac{2A}{3\pi} = \sqrt{\frac{2A}{3\pi}} \left(-\pi^2 \frac{2A}{3} + \frac{\pi}{\cancel{2}} 4\pi \frac{2A}{3} + 4\pi^2 \frac{2A}{3} \right) =$$

$$= \sqrt{\frac{2A}{3\pi}} \left(-\frac{2}{3} \pi^2 A + \frac{4}{3} \pi^2 A + \frac{8}{3} \pi^2 A \right) = \sqrt{\frac{2A}{3\pi}} \frac{10}{3} \pi^2 A > 0$$

máximo
condición necesario

El volumen de un cilindro de área de superficie A será máximo si su diámetro es $\sqrt{\frac{2A}{3\pi}}$ y su altura es $\sqrt{\frac{2A}{3\pi}}$

$$b) L = 2\pi rh + 2\pi r^2 + \lambda(\pi r^2 h - V)$$

$$L'_r = 2\pi h + 4\pi r + 2\lambda\pi rh = 0 \quad (1)$$

$$L'_h = 2\pi r + \lambda\pi r^2 = 0 \quad (2)$$

$$L'_\lambda = \pi r^2 h - V = 0 \quad (3)$$

$$\text{De (2)} \rightarrow \pi r(2 + \lambda r) = 0 \Rightarrow \begin{cases} r = 0 & (\text{absurdo; es en radio}) \\ \lambda r = -2 \Rightarrow \lambda = -\frac{2}{r} & (r \neq 0) \rightarrow \frac{2}{r} \end{cases}$$

$$\text{De (3)} \rightarrow h = \frac{V}{\pi r^2} \quad (\text{que } r \neq 0 \text{ por ser un radio})$$

$$\text{Reemplazando en (1)} \rightarrow 2\pi \frac{V}{\pi r^2} + 4\pi r + 2\left(-\frac{2}{r}\right)\pi \frac{V}{\pi r^2} = 0$$

$$\Rightarrow \frac{2V}{r^2} + 4\pi r - \frac{4V}{r^2} = 0 \Rightarrow 4\pi r - \frac{2V}{r^2} = 0 \Rightarrow$$

$$\Rightarrow \frac{4\pi r^3 - 2V}{r^2} = 0 \Rightarrow 4\pi r^3 = 2V \Rightarrow r = \sqrt[3]{\frac{2V}{4\pi}} \Rightarrow$$

$$\Rightarrow r = \sqrt[3]{\frac{V}{2\pi}}$$

$$P = \left(\sqrt[3]{\frac{V}{2\pi}}, \frac{V}{\pi \left(\frac{V}{2\pi}\right)^{2/3}} \right) \text{ con } \lambda = \frac{-2}{\sqrt[3]{\frac{V}{2\pi}}}$$

$$P = \left[\left(\frac{V}{2\pi}\right)^{1/3}, \left(\frac{V}{\pi}\right)^{1/3} \sqrt[3]{4} \right] \text{ con } \lambda = -2\left(\frac{2\pi}{V}\right)^{1/3}$$

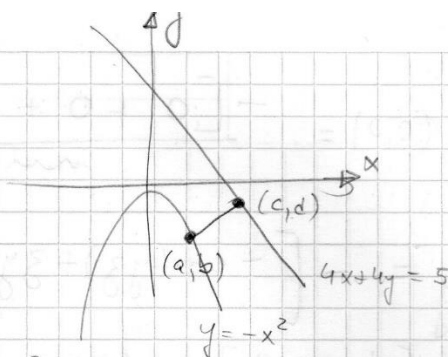
Por el tipo de problema analizado, P debe ser un mínimo; por lo tanto: para un volumen dado V , el área del cilindro será mínima si el diámetro

$$\text{es } \underbrace{2\left(\frac{V}{2\pi}\right)^{1/3}}_{\left(\frac{4V}{\pi}\right)^{1/3}} \text{ y la altura } \underbrace{\sqrt[3]{4} \left(\frac{V}{\pi}\right)^{1/3}}_{\left(\frac{4V}{\pi}\right)^{1/3}}$$

$$20) \quad d = \sqrt{(a-c)^2 + (b-d)^2}$$

Pto parabola $(a, -a^2)$

Pto recta $(x, \frac{5-4x}{4})$



Análisis mínimo de d^2

$$g(a, x) = (x-a)^2 + \left(\frac{5-4x}{4} + a^2\right)^2$$

$$\left\{ \begin{array}{l} g'_x = 2(x-a) + 2\left(\frac{5-4x}{4} + a^2\right) \cdot (-1) = 0 \quad (1) \\ g'_a = -2(x-a) + 2\left(\frac{5-4x}{4} + a^2\right) \cdot 2a = 0 \quad (2) \end{array} \right.$$

$$(1) + (2) \Rightarrow 0 = -2\left(\frac{5-4x}{4} + a^2\right) + 4a\left(\frac{5-4x}{4} + a^2\right) = 0 \Rightarrow$$

$$\Rightarrow \underbrace{-2 + 4a = 0}_a = \frac{1}{2} \quad \vee \quad \underbrace{\frac{5-4x}{4} + a^2 = 0}_{\frac{5+4a^2}{4} = x}$$

$$5-4x = -4a^2$$

$$\frac{5+4a^2}{4} = x$$

$$\frac{1}{2} = a$$

Reemp en (1)

$$2\left(x - \frac{1}{2}\right) - 2\left(\frac{5-4x}{4} + \frac{1}{4}\right) = 0 \Rightarrow$$

$$\Rightarrow -x - \frac{1}{2} - \frac{5}{4} + x - \frac{1}{4} = 0 \Rightarrow 2x^2 = \frac{3}{2} + \frac{1}{2} \Rightarrow$$

$$\Rightarrow 2x = 2 \Rightarrow x = 1 \quad \text{el pto crítico es } \rightarrow P_1 = \left(\frac{1}{2}, 1\right)$$

$$x = \frac{5+4a^2}{4} \quad \text{Reemp en (1)}$$

$$2\left(\frac{5+4a^2}{4} - a\right) - 2\left(\frac{5-4\left(\frac{5+4a^2}{4}\right)}{4} + a^2\right) = 0$$

$$\Rightarrow \cancel{\frac{5}{4}} + \cancel{a^2} - a - \cancel{\frac{5}{4}} + \frac{5+4a^2}{4} + \cancel{a^2} = 0 \Rightarrow$$

$$\Rightarrow -a + \frac{5}{4} + a^2 = 0$$

$$a_{1,2} = \frac{1 \pm \sqrt{1 - 4 \cdot \frac{5}{4}}}{2} \notin \mathbb{R} \quad !!$$

El único pto crítico es mínimo

$$a = \frac{1}{2} \rightarrow \text{Pto de la parábola } \left(\frac{1}{2}, -\frac{1}{4} \right)$$

$$x = 1 \rightarrow \text{Pto de la recta } \left(1, \frac{1}{4} \right)$$

Distancia entre ambos:

$$d = \sqrt{\left(\frac{1}{2} - 1 \right)^2 + \left(-\frac{1}{4} - \frac{1}{4} \right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \left(\frac{\sqrt{2}}{2} \right)$$